Graphical Quadratic Algebra

Fabio Zanasi

UCL

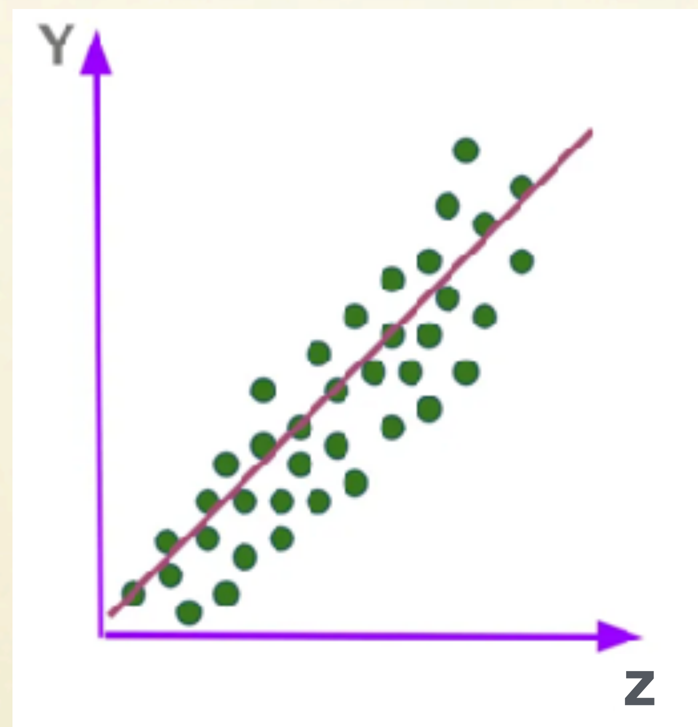
Joint work with

Dario Stein   Richard Samuelson   Robin Piedeleu

Introduction

What we want to model

Dependent variable Y



Independent variable Z
(regressor)

Observations
(in green)

Goal of linear regression:
to find a linear model that best fits a set of observations

What we want to model

Linear algebraic perspective

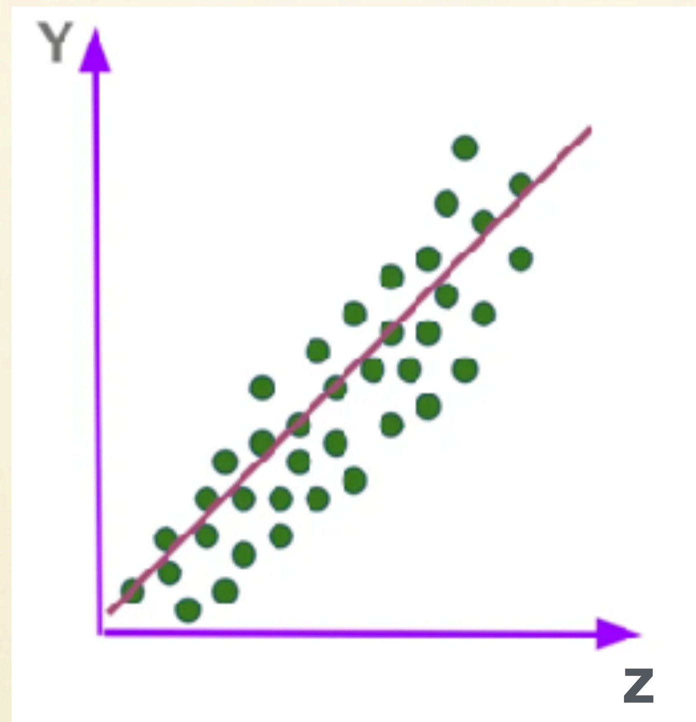
$$Ax = b$$

minimise values of

$$f(x) = \frac{1}{2}\|Ax - b\|^2$$

theory of partial quadratic functions

find optimal x as the \`ordinary least squares estimator' (OLSE)



If $\epsilon \sim \mathcal{N}(0,\Sigma)$
then OLSE =
maximum likelihood

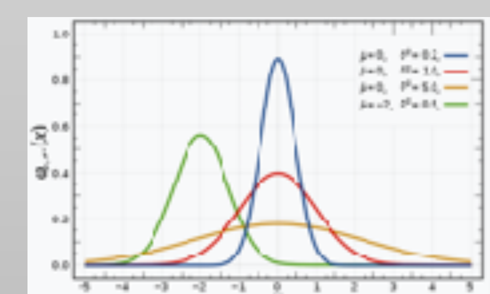
Stochastic perspective

random variables
x (parameters), ϵ (error)

maximise
the likelihood of

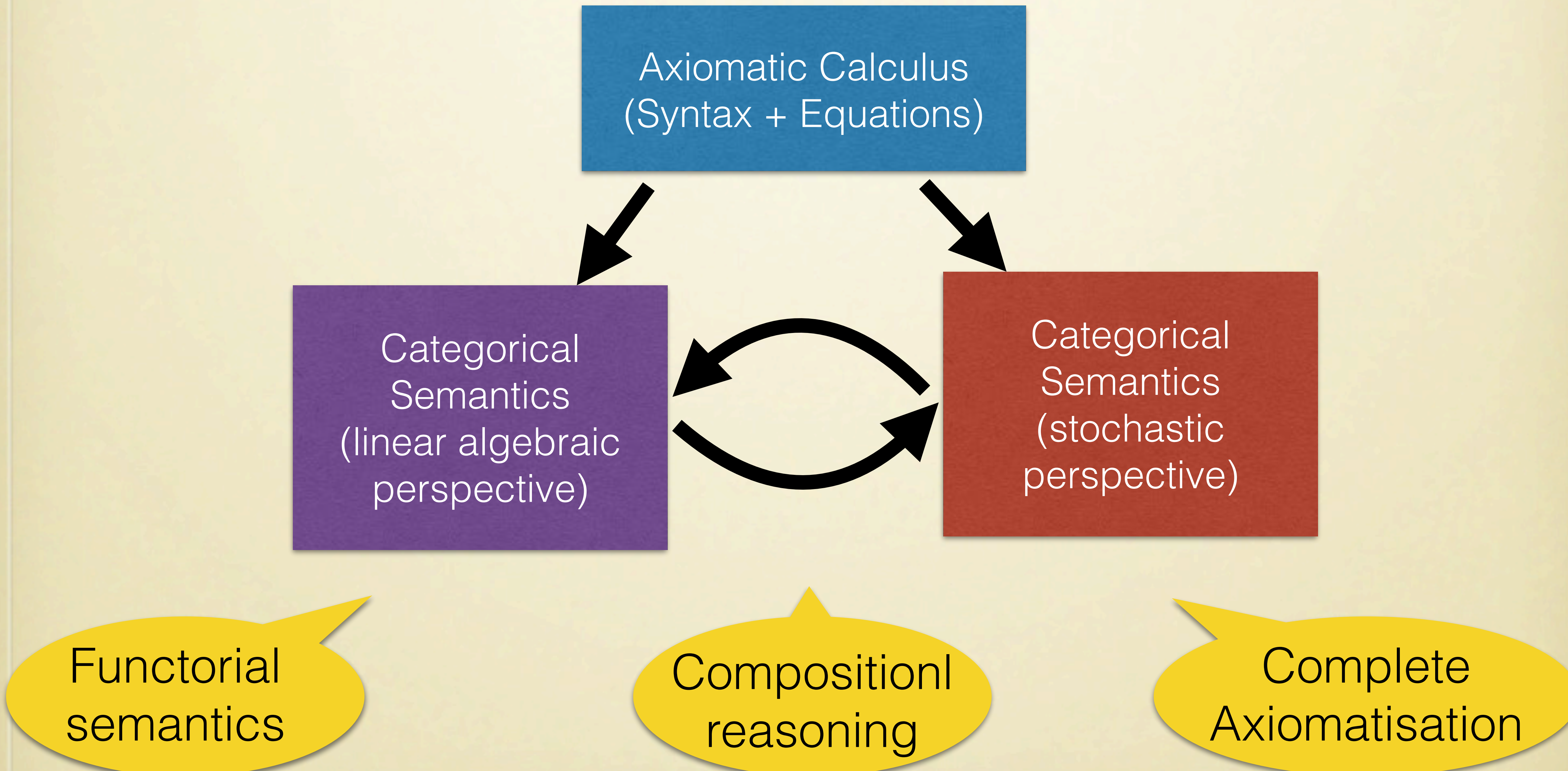
$$Ax + \epsilon = b$$

theory of Gaussian distributions



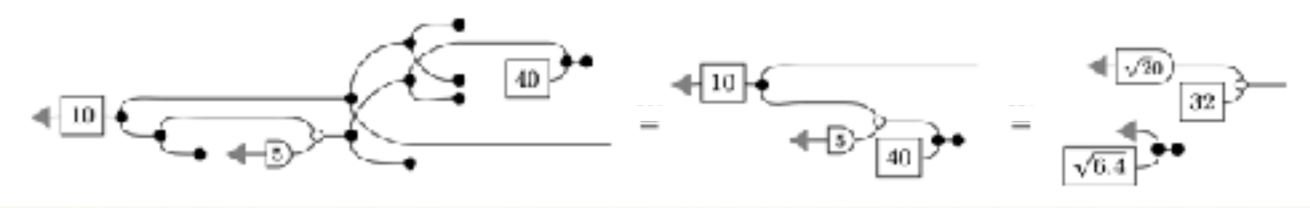
Aims

A **categorical model** of quadratic problems



Methodology

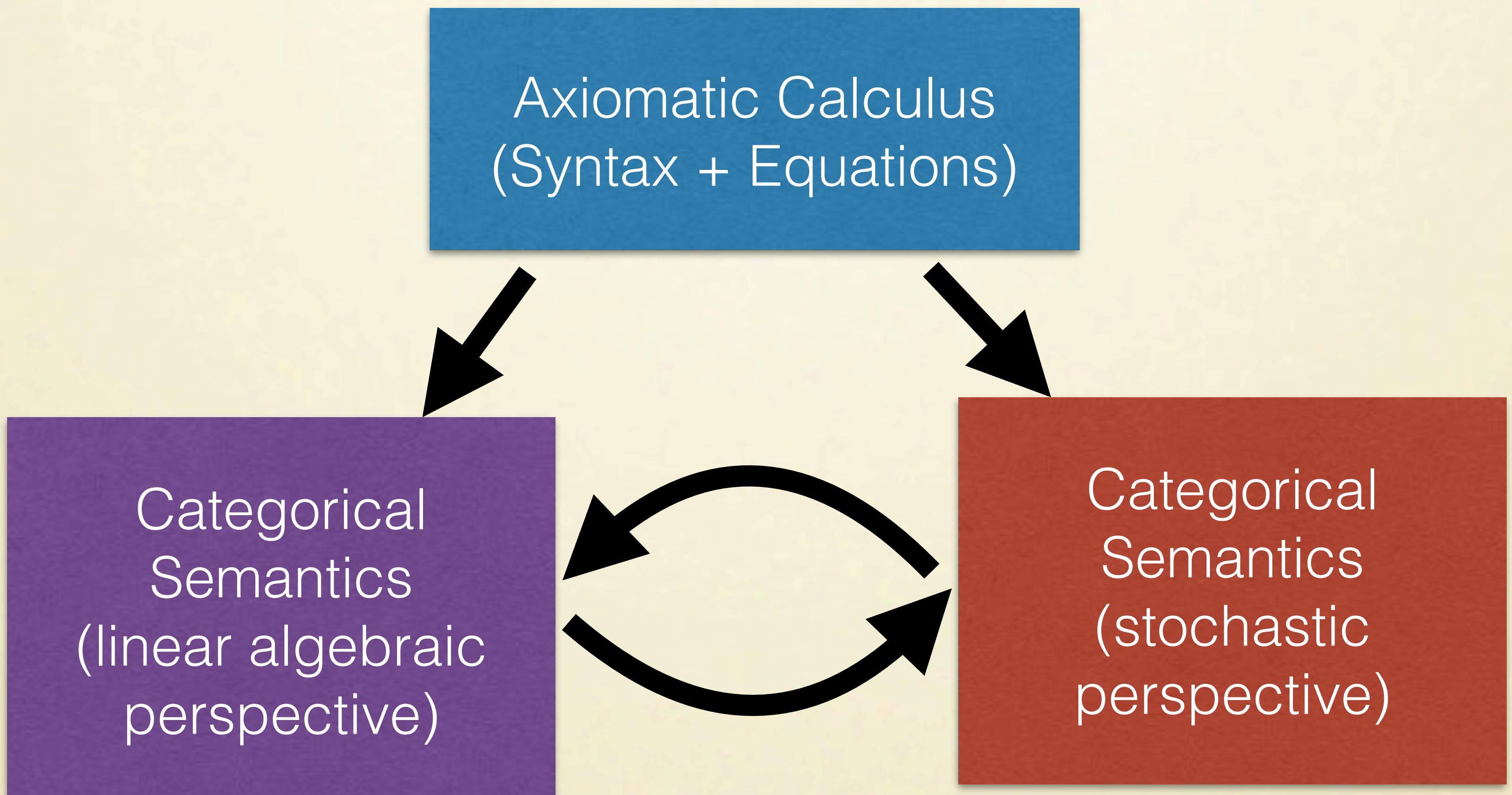
The syntax of our calculus will be based on **string diagrams**.



- Expressive Formalism for Resource Handling (copy/discard).
- Build on existing diagrammatic calculus for affine spaces.
- Connect to large body of work of string diagrams for dynamical systems, probabilistic programming, quantum theory.

Axiomatising Gaussian Processes

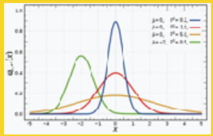
Aims


Axiomatic Calculus
(Syntax + Equations)
Categorical Semantics (linear algebraic perspective)
Categorical Semantics (stochastic perspective)

Aims

Normal distributions

$\mathcal{N}(\mu, \Sigma)$



Multivariate Normal Distribution

specifiable by $\mathcal{N}(\mu, \Sigma)$
and an affine map.

Axiomatic Calc
(Syntax + Equa

Categorical
Semantics
(linear algebraic
perspective)

Categorical
Semantics
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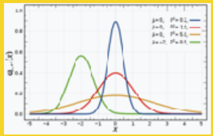

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Axiomatic Calc
(Syntax + Equa

Categorical
Semantics
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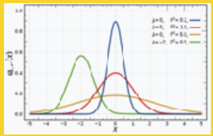
Gauss

objects: $\mathbb{R}^n$
morphisms are multivariate
normal distributions
$f(x) = Ax + \mathcal{N}(\mu, \Sigma)$

Aims

Normal distributions

$\mathcal{N}(\mu, \Sigma)$



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objects: $\mathbb{R}^n$
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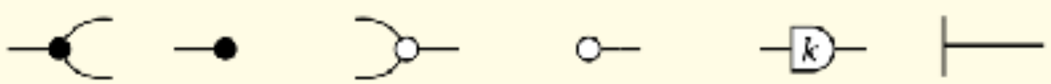
$f(x) = Ax + \mathcal{N}(\mu, \Sigma)$

Normal distributions
are maps $0 \to \mathbb{R}^n$

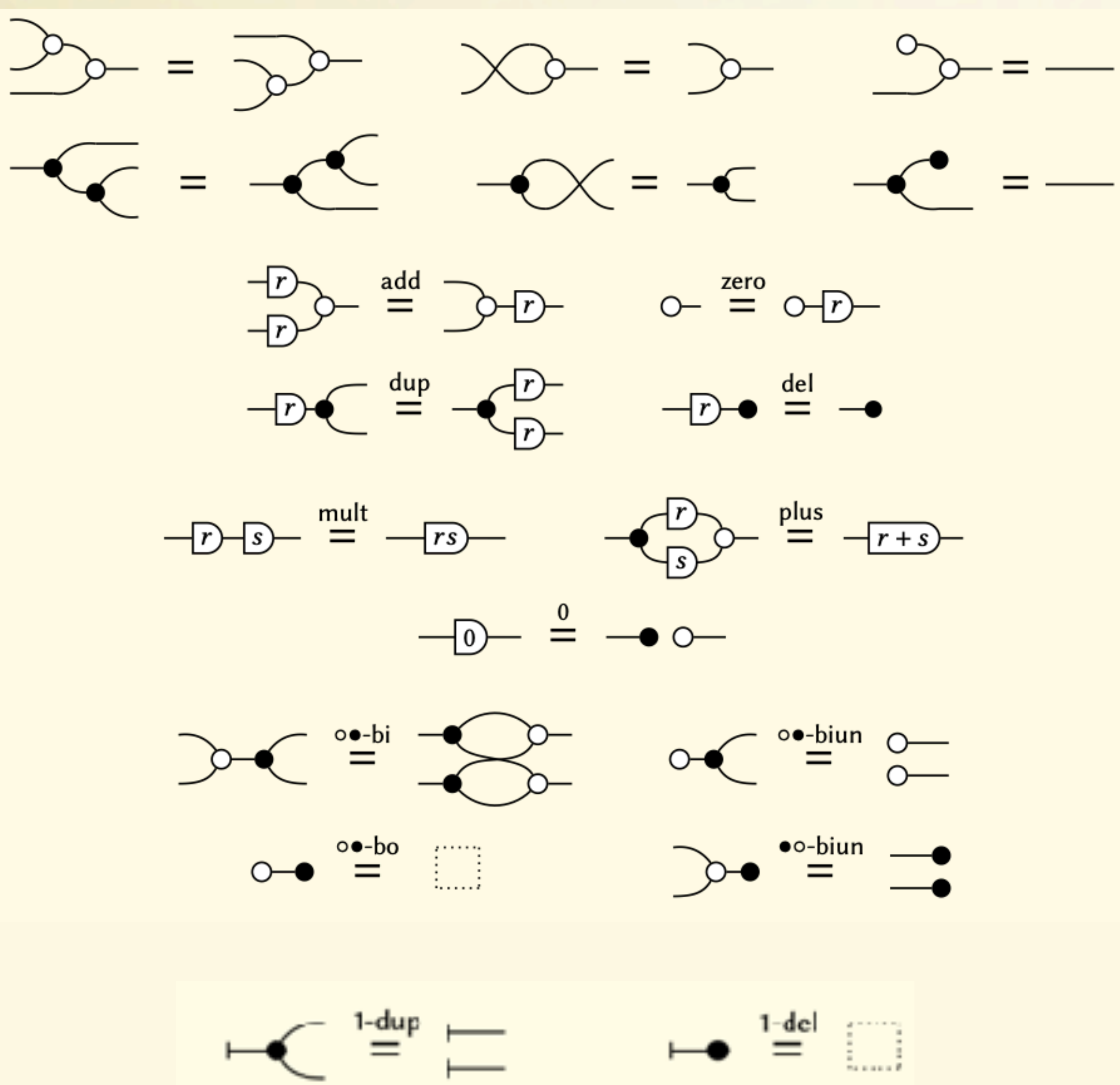
Starting Point: Graphical Affine Algebra

LICS'19

copy discard sum zero scale one



axiomatisation complete for affine maps



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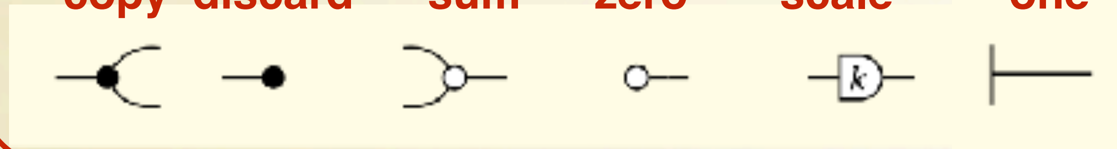

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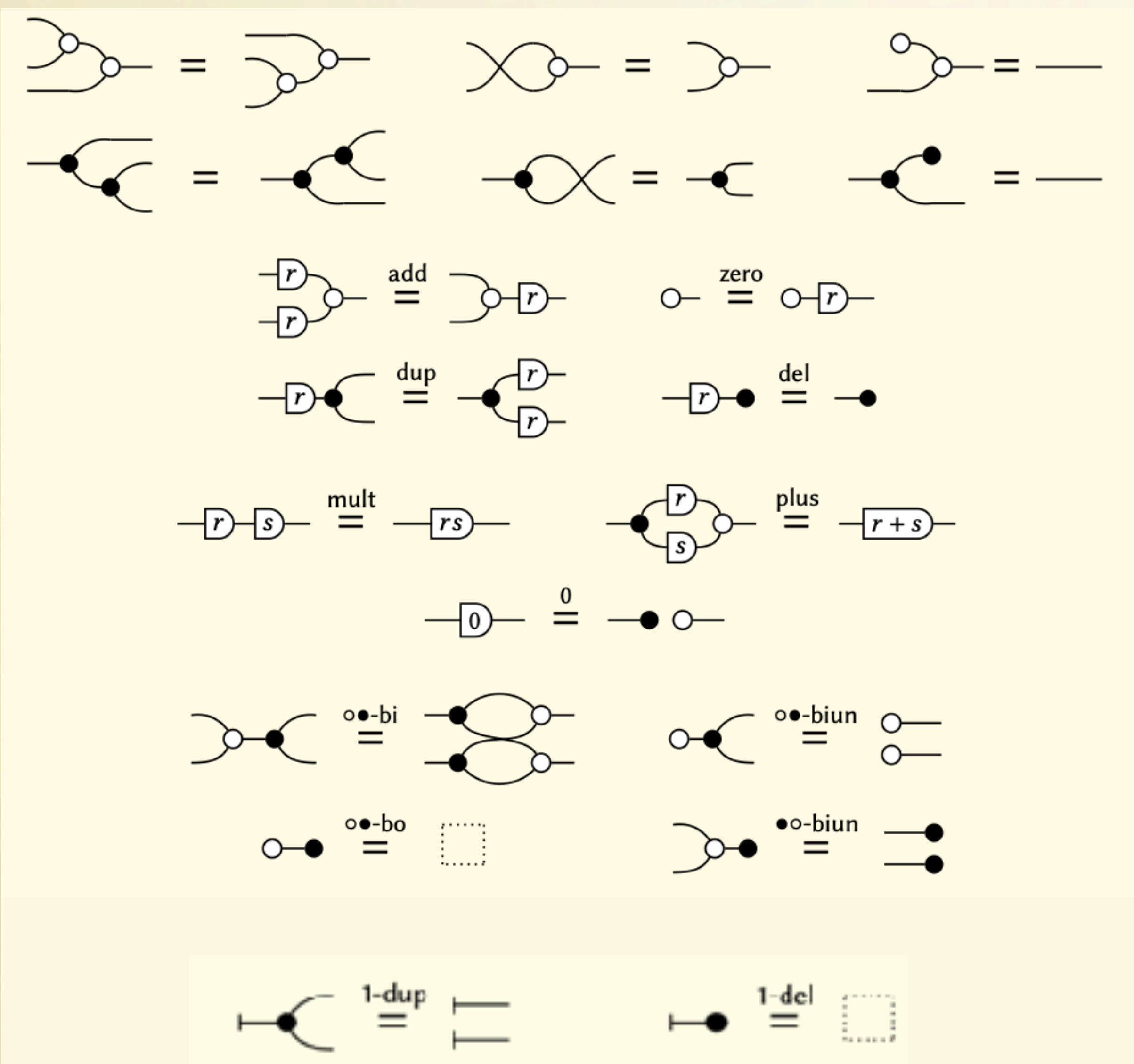
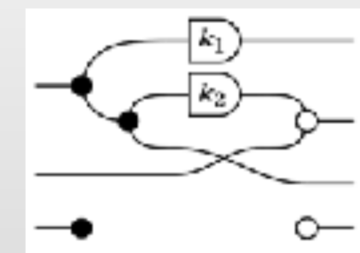


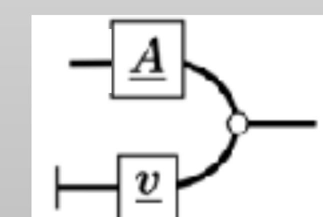
Picturing matrices with coefficients in a field

$$\begin{pmatrix} k_1 & 0 & 0 \\ k_2 & 0 & 1 \\ 1 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix} \longmapsto$$



Picturing affine maps

$$f(x) = Ax + v \longmapsto$$



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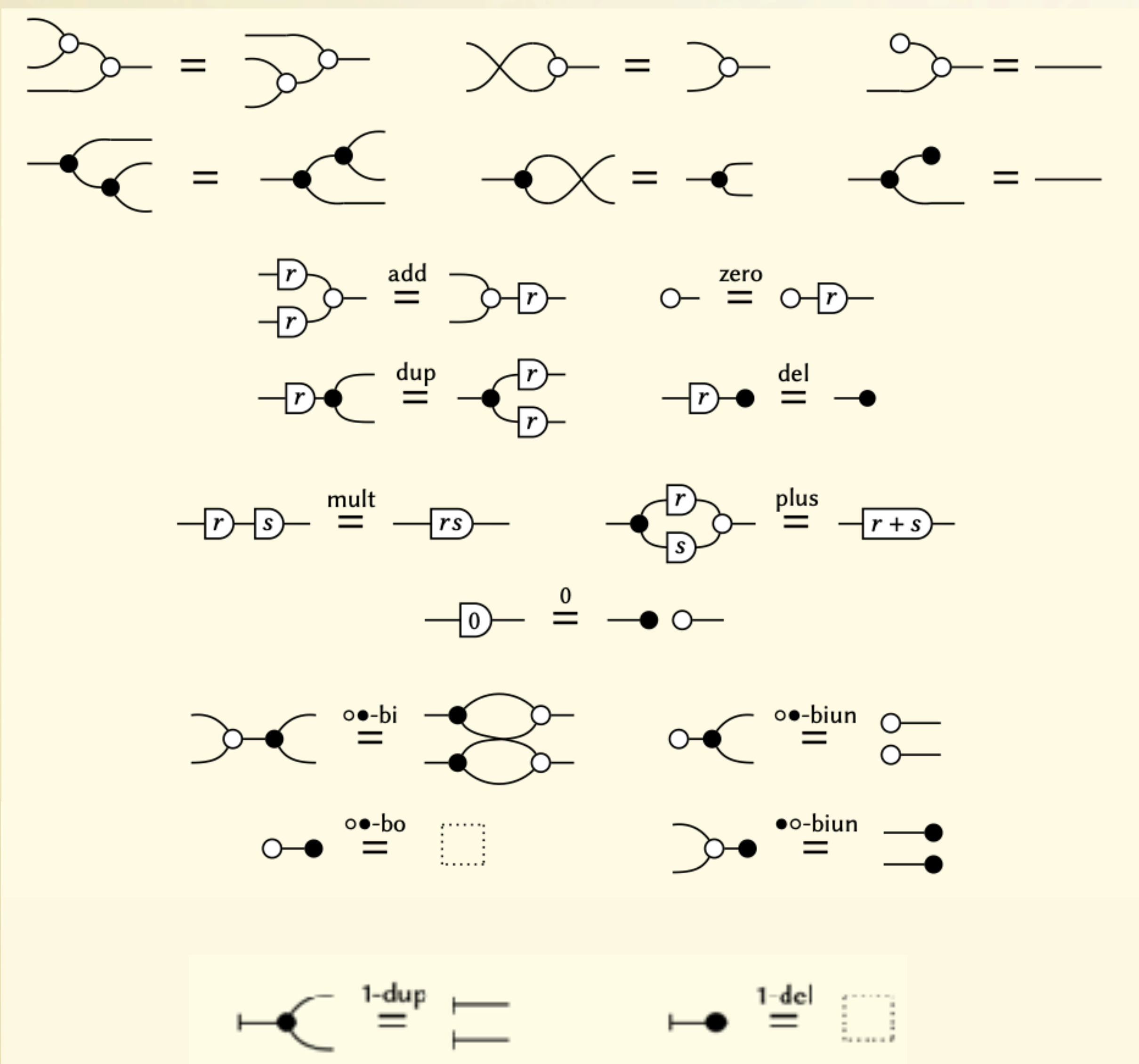

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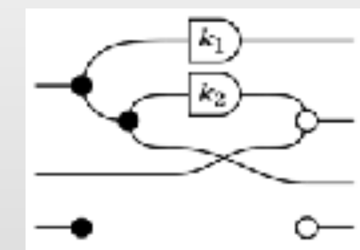

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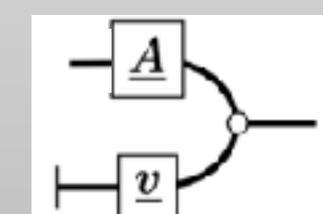
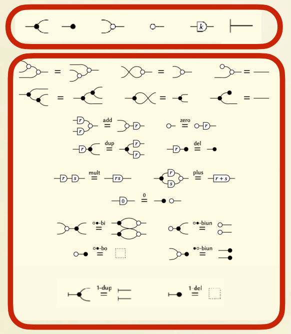


How do we extend to Gaussian probability?

Graphical Quadratic Algebra: Syntax & Semantics

GAA



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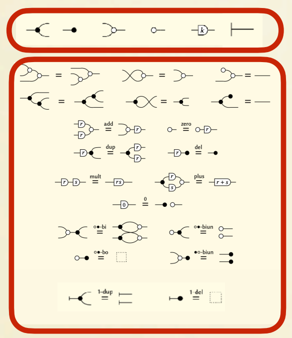
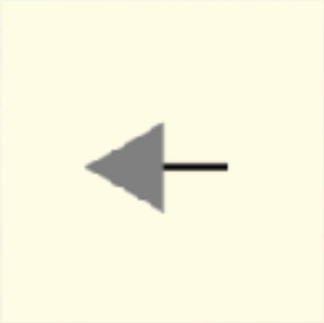


+

Extra generator



Graphical Quadratic Algebra: Syntax & Semantics

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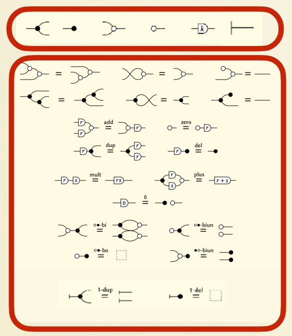


+

Extra generator

In **Gauss**

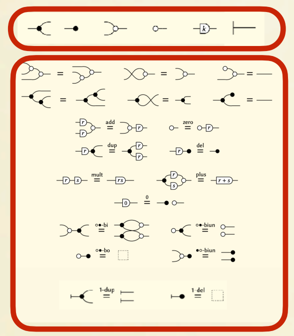
The standard

Gaussian distribution

$\mathcal{N}(0,1)$

Graphical Quadratic Algebra: Syntax & Semantics

GAA



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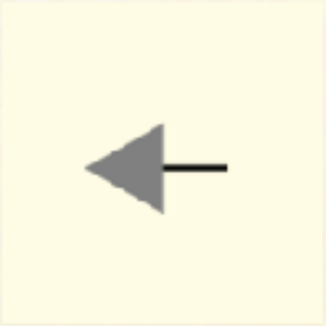
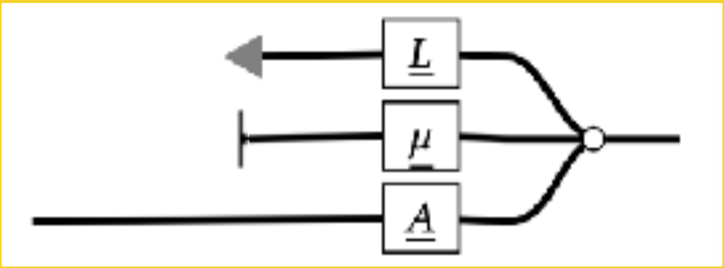
Extra generator



In **Gauss**

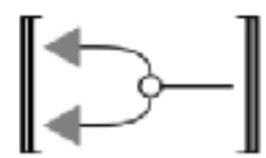
Representation of a **Gauss-**morphism $f(x) = Ax + \mathcal{N}(\mu, \Sigma)$.



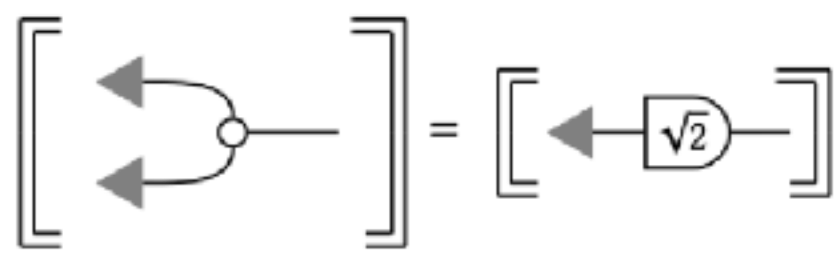
with $\Sigma = LL^T$

The standard Gaussian distribution $\mathcal{N}(0,1)$

Graphical Quadratic Algebra: Figuring out the equations

 $= \mathcal{N}(0,2)$

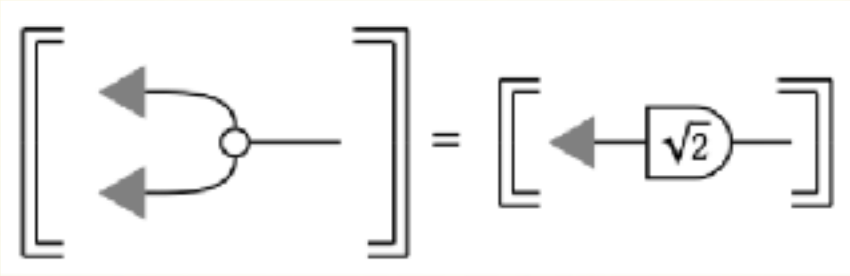
In **Gauss**

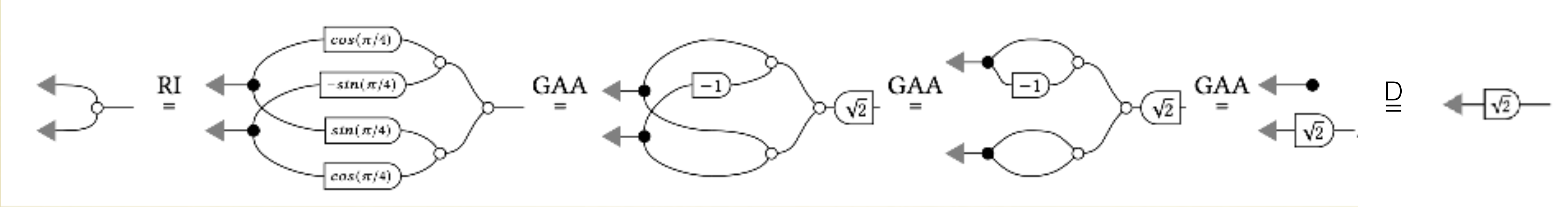


Reading: if $X_1, X_2 \sim \mathcal{N}(0,1)$ are independent random variables, the $X_1 + X_2$ is distributed according to $\mathcal{N}(0,2)$.

What equations do we need to prove this fact?

Graphical Quadratic Algebra: Figuring out the equations





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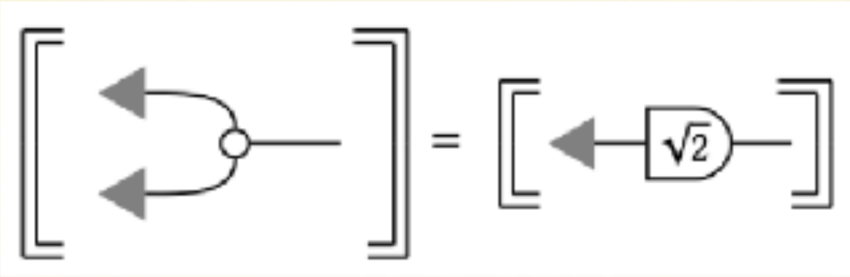

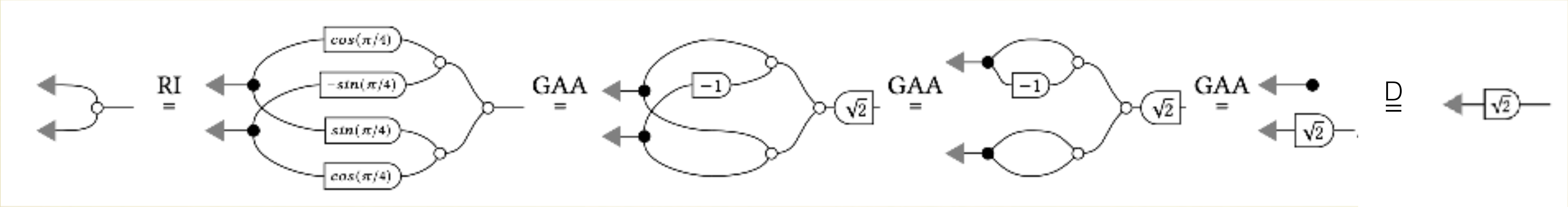

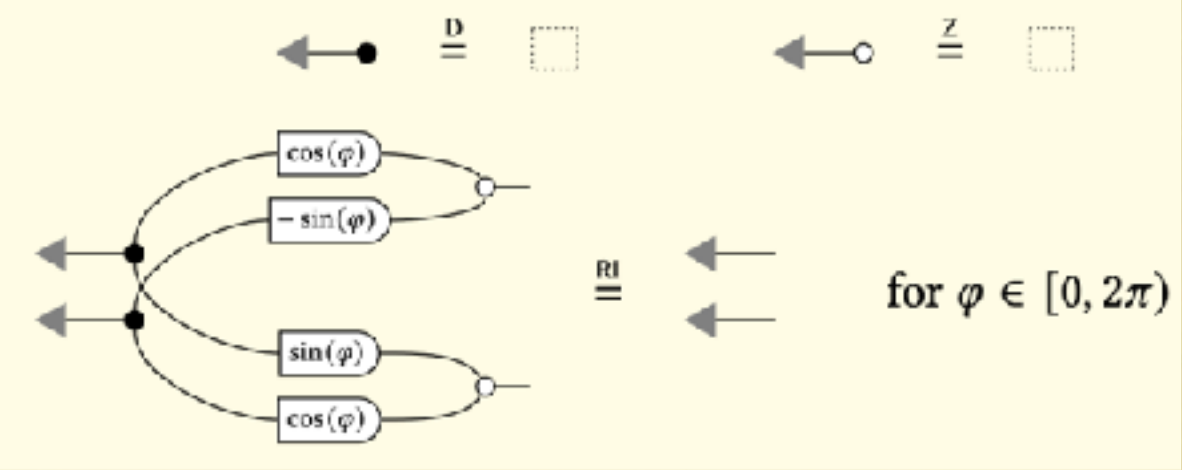

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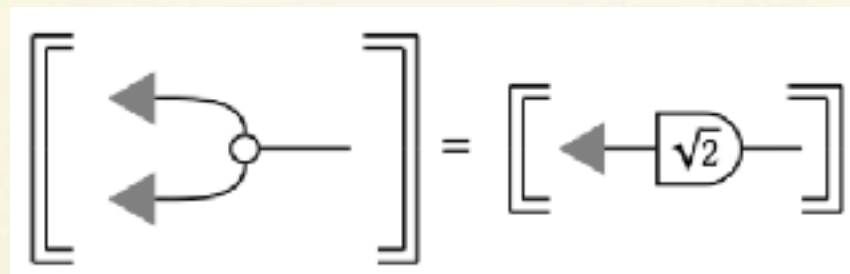
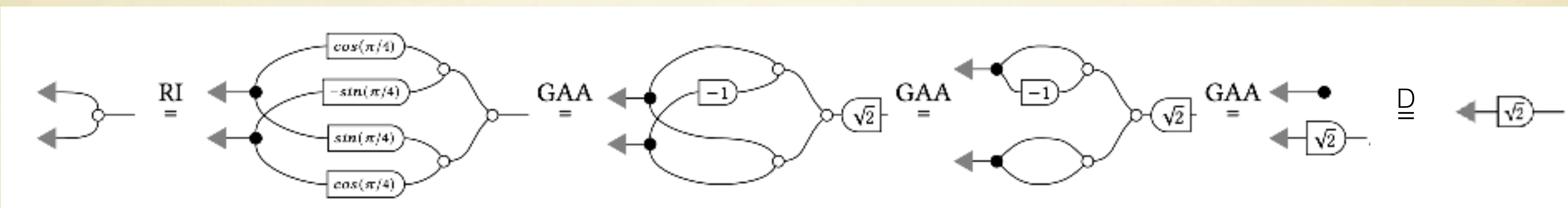
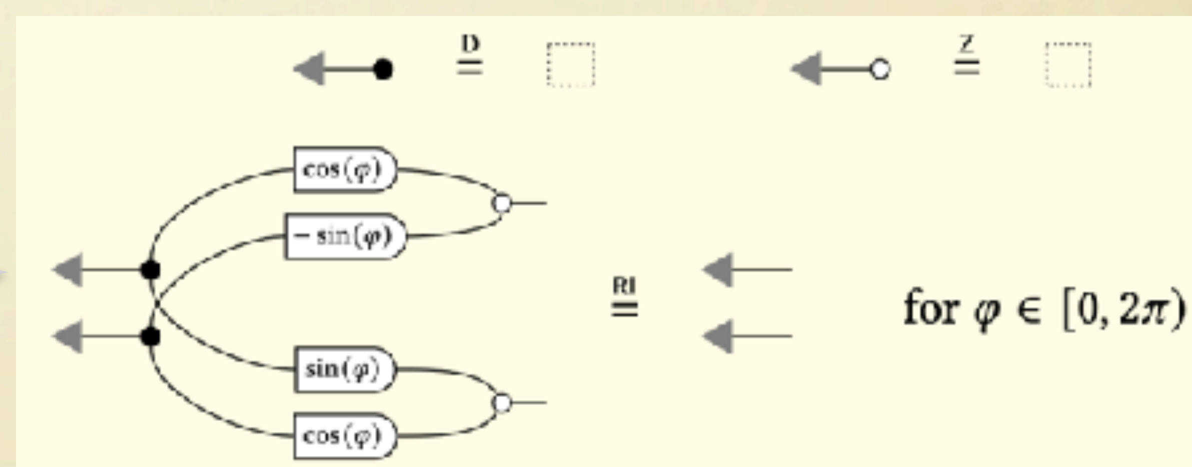


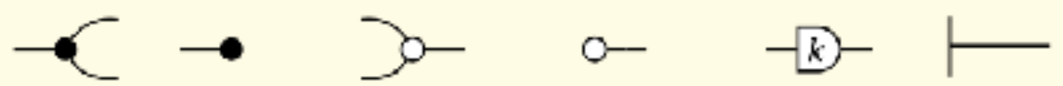


Key derived law:  is invariant under any orthogonal matrix (reminiscent of the Herschel-Maxwell theorem!)

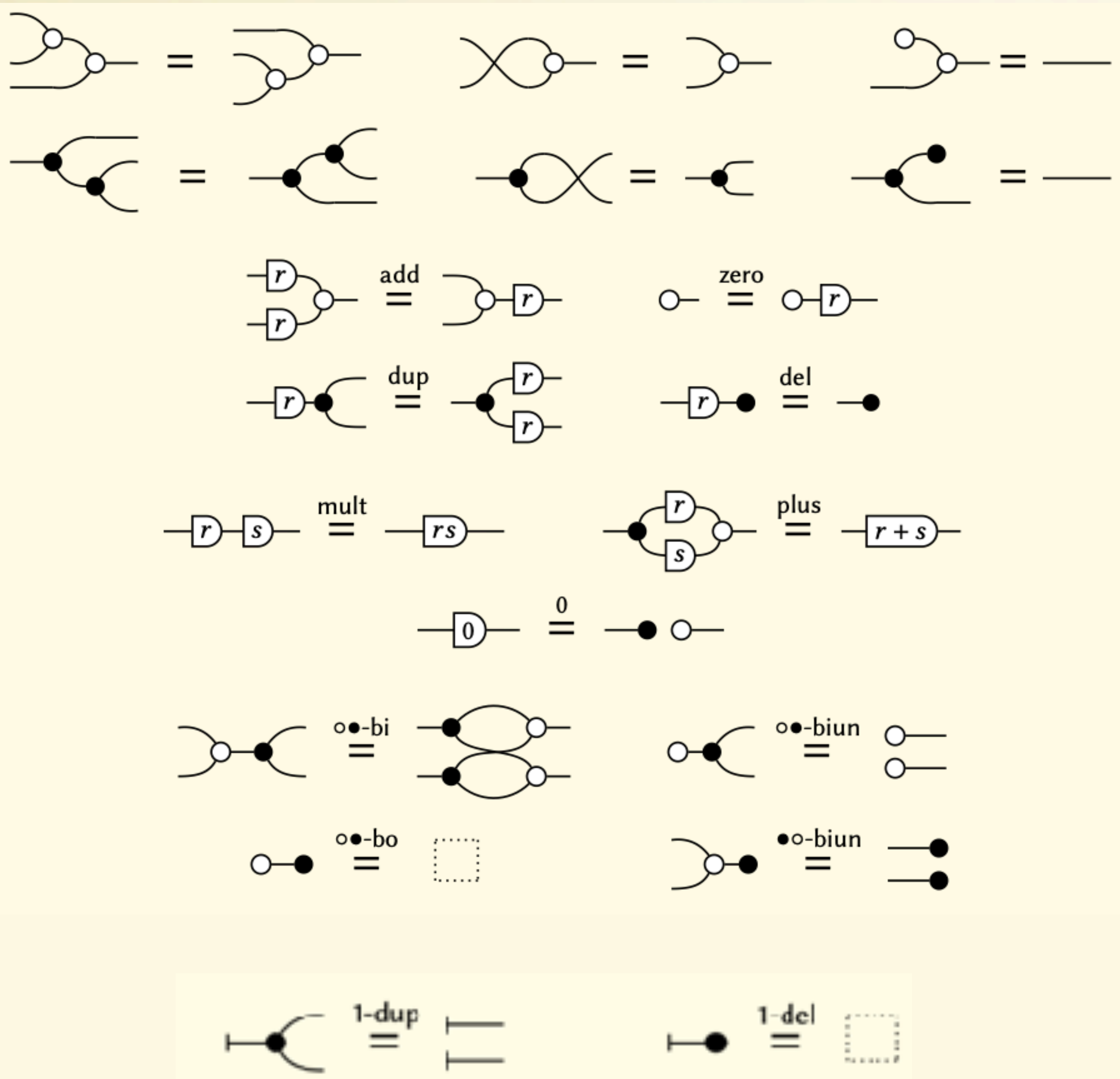


Graphical Quadratic Algebra

copy discard sum zero scale one



axiomatisation complete for affine maps



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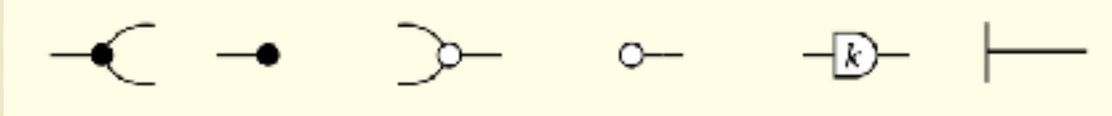

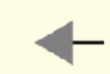

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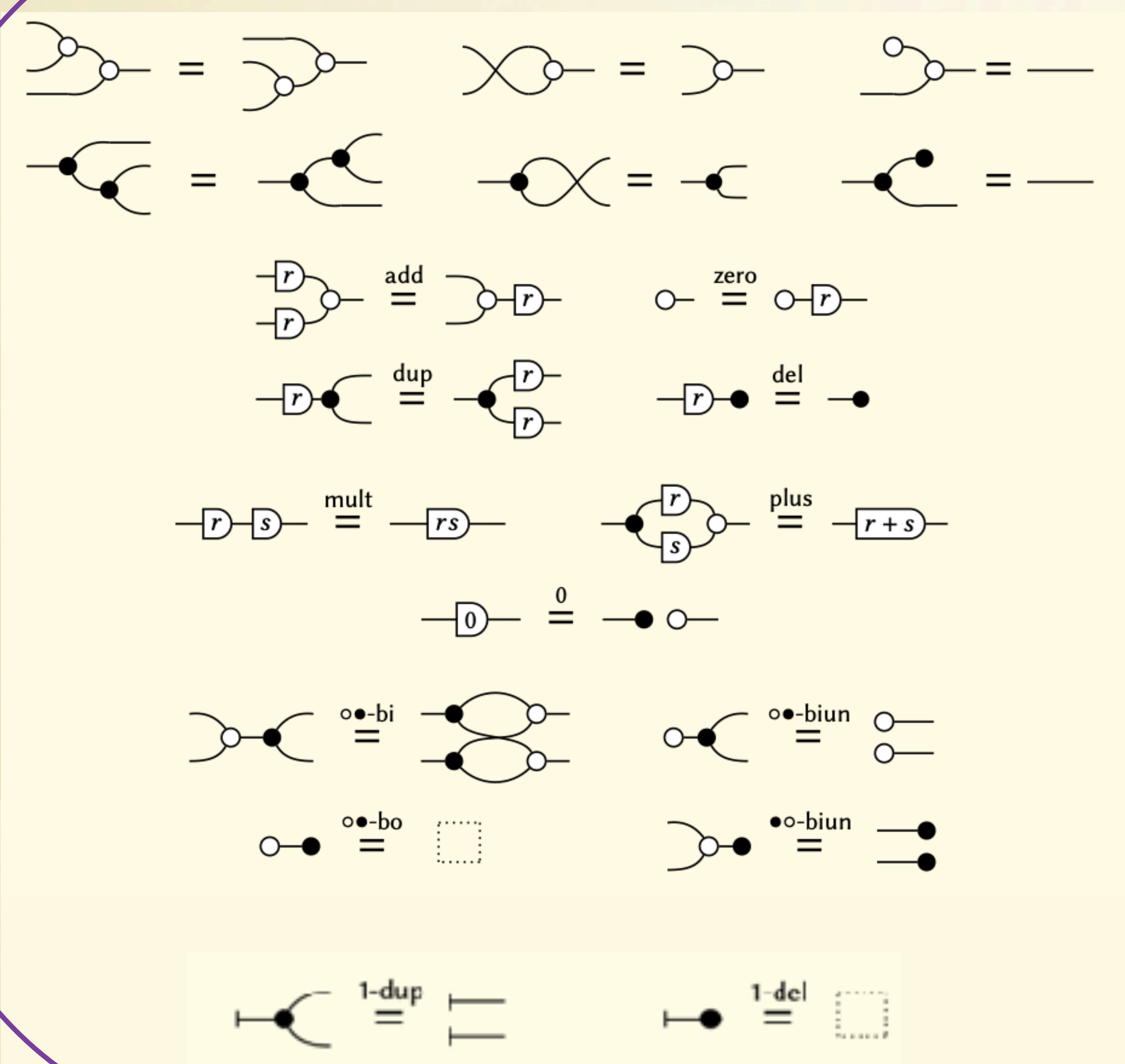

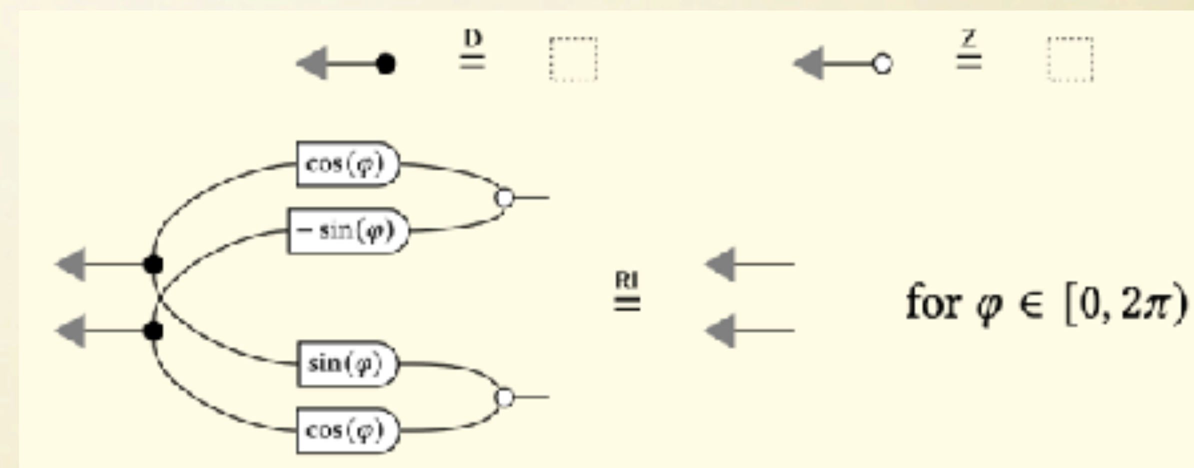

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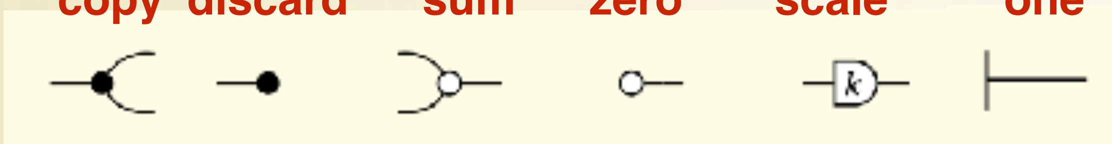

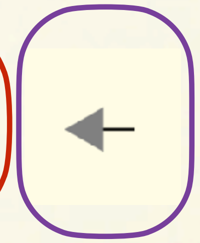

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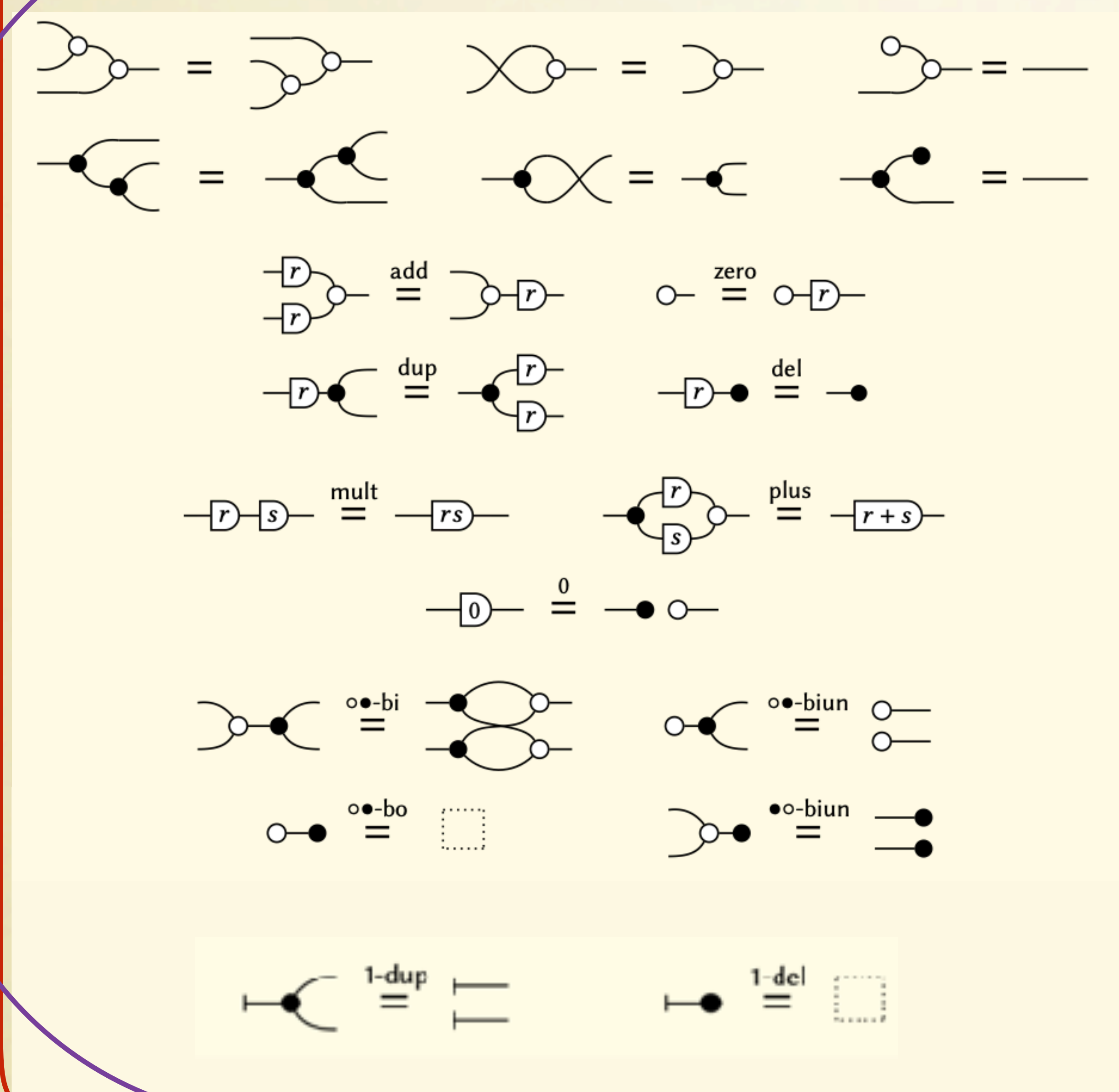
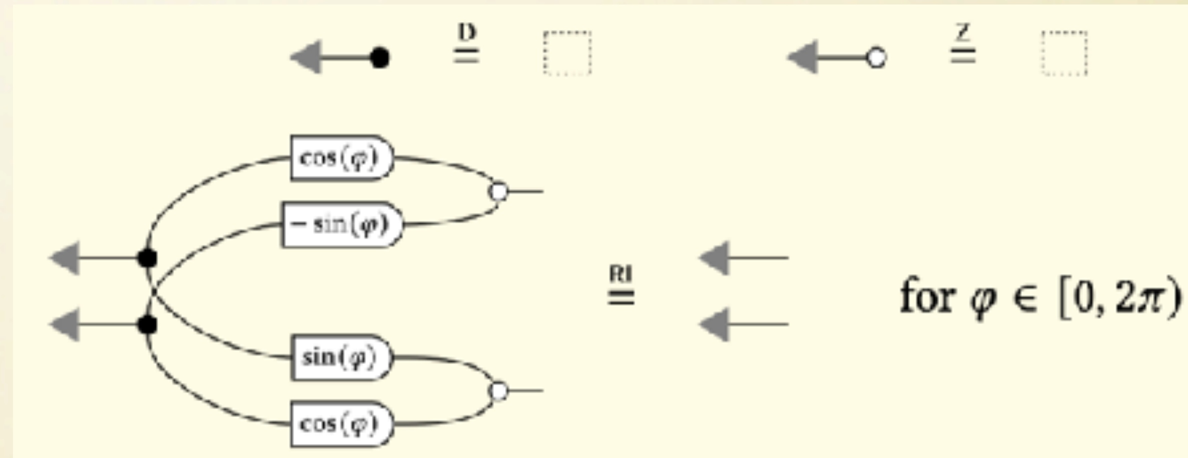
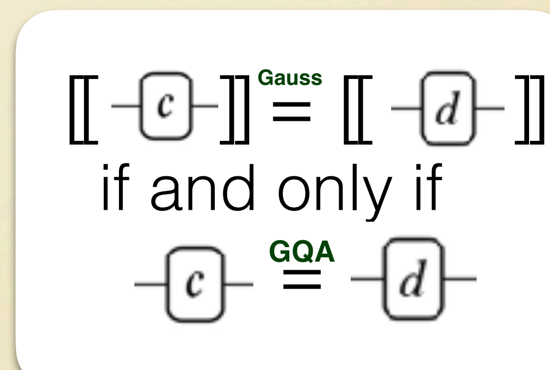




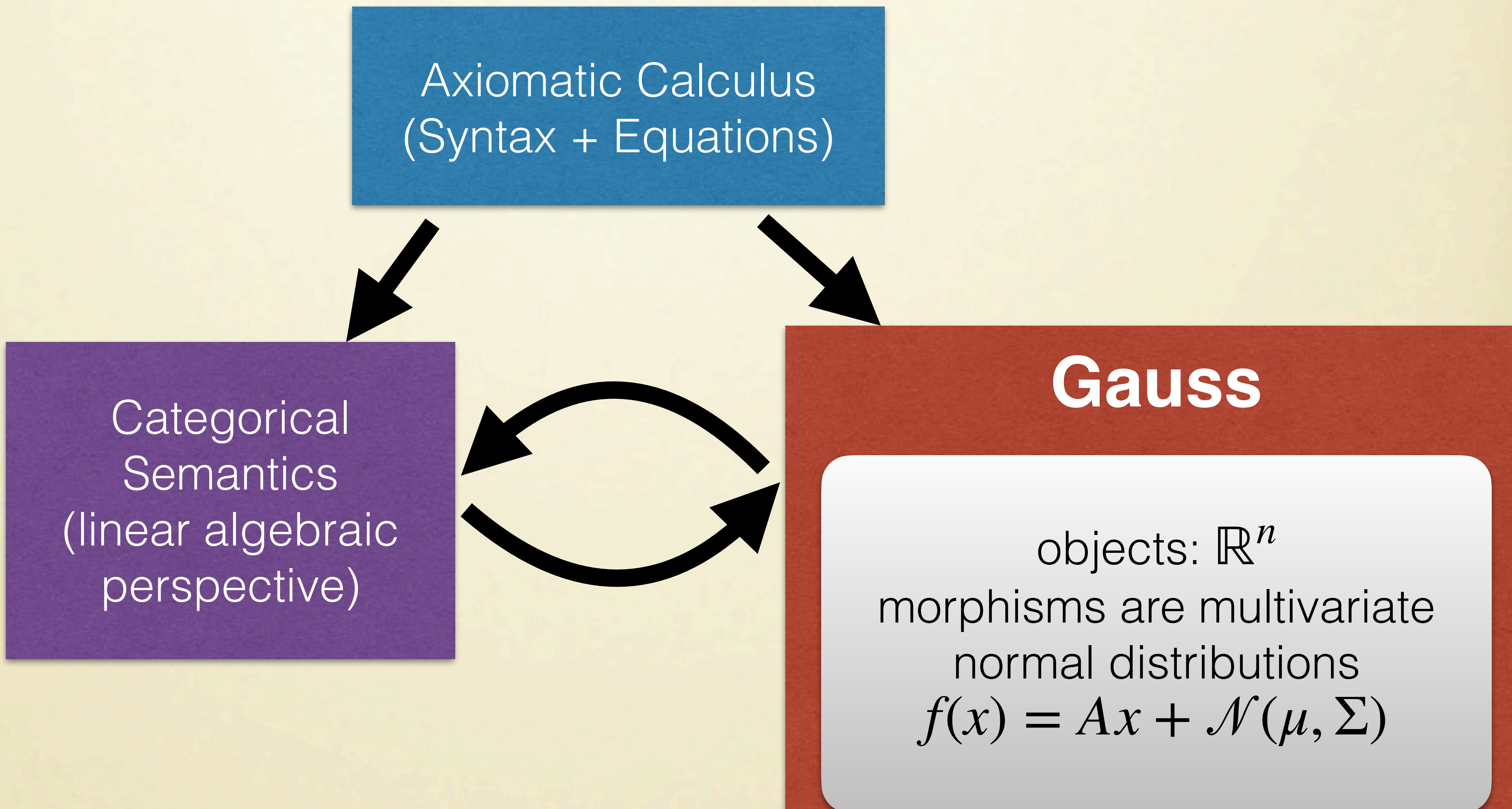
Theorem

This axiomatisation is complete for Gaussian maps.



Axiomatising
Quadratic Problems

Aims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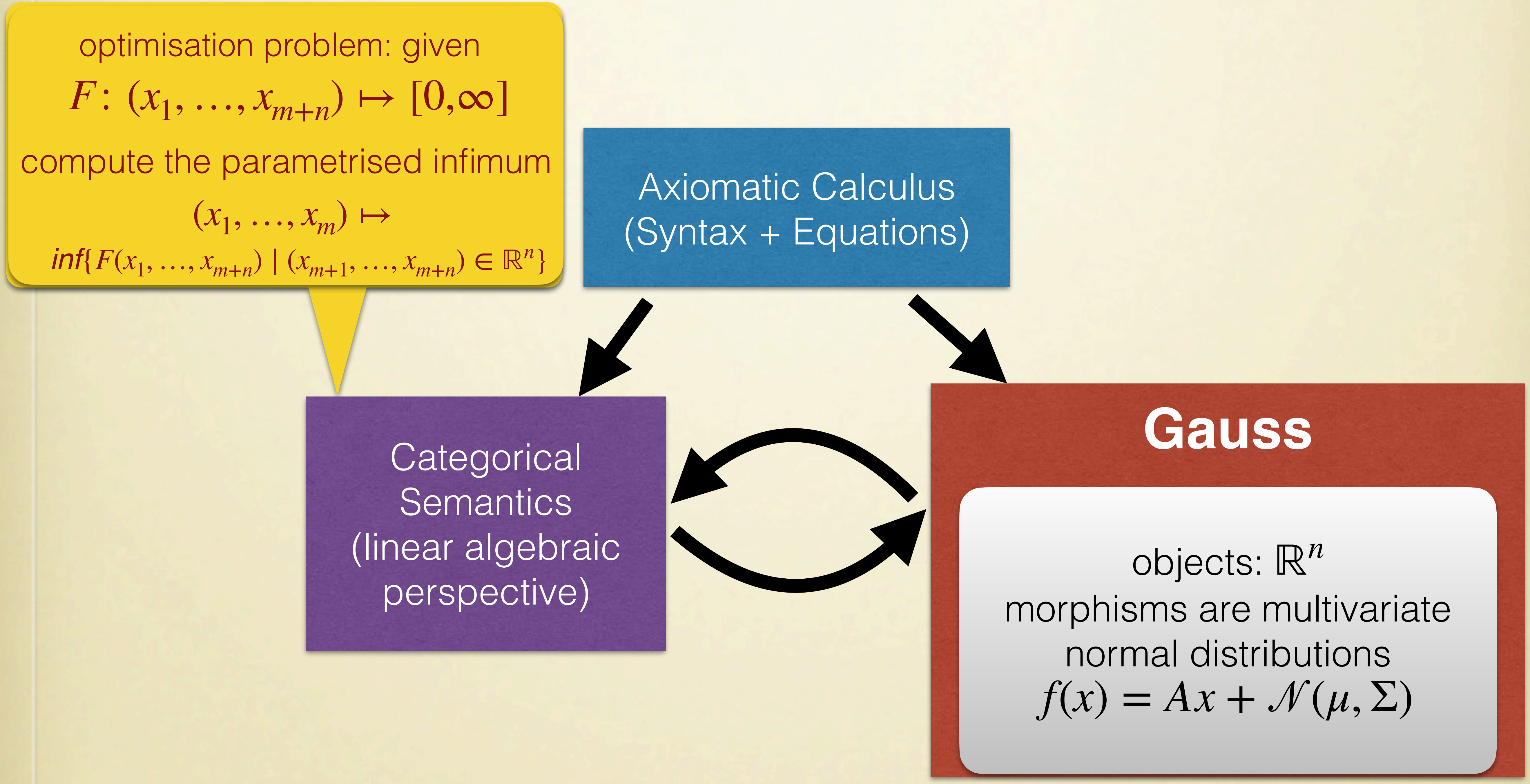
optimisation problem: given
$F: (x_1, \dots, x_{m+n}) \mapsto [0,\infty]$
compute the parametrised infimum
$(x_1, \dots, x_m) \mapsto$
inf$\{F(x_1, \dots, x_{m+n}) \mid (x_{m+1}, \dots, x_{m+n}) \in \mathbb{R}^n\}$
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optimisation problem: given

$F\colon (x_1, \ldots, x_{m+n}) \mapsto [0,\infty]$

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inf$\{F(x_1, \ldots, x_{m+n}) \mid (x_{m+1}, \ldots, x_{m+n}) \in \mathbb{R}^n\}$

Axiomatic Calculus
(Syntax + Equations)

Quadrel

objects: $\mathbb{R}^n$

morphisms $\mathbb{R}^n \to \mathbb{R}^m$ are non-negative partial quadratic functions $f\colon \mathbb{R}^n \times \mathbb{R}^m \to [0,\infty]$

which compose relationally:

$$(f;g)(x,z) = \inf_y \{f(x,y) + g(y,z)\}$$

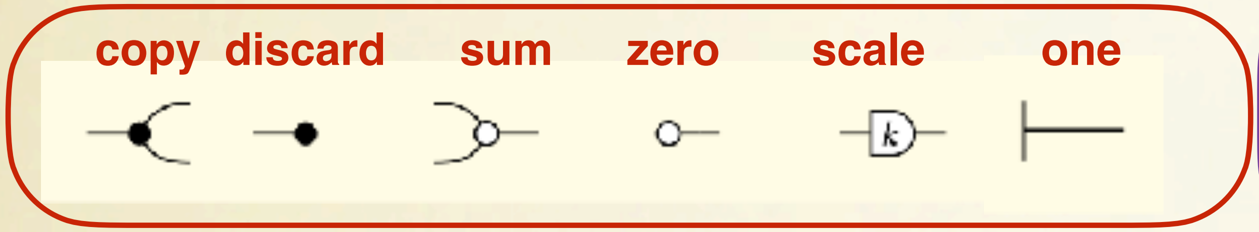
Gauss

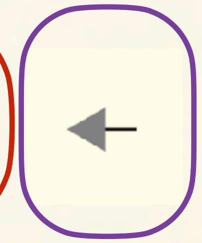
objects: $\mathbb{R}^n$

morphisms are multivariate normal distributions

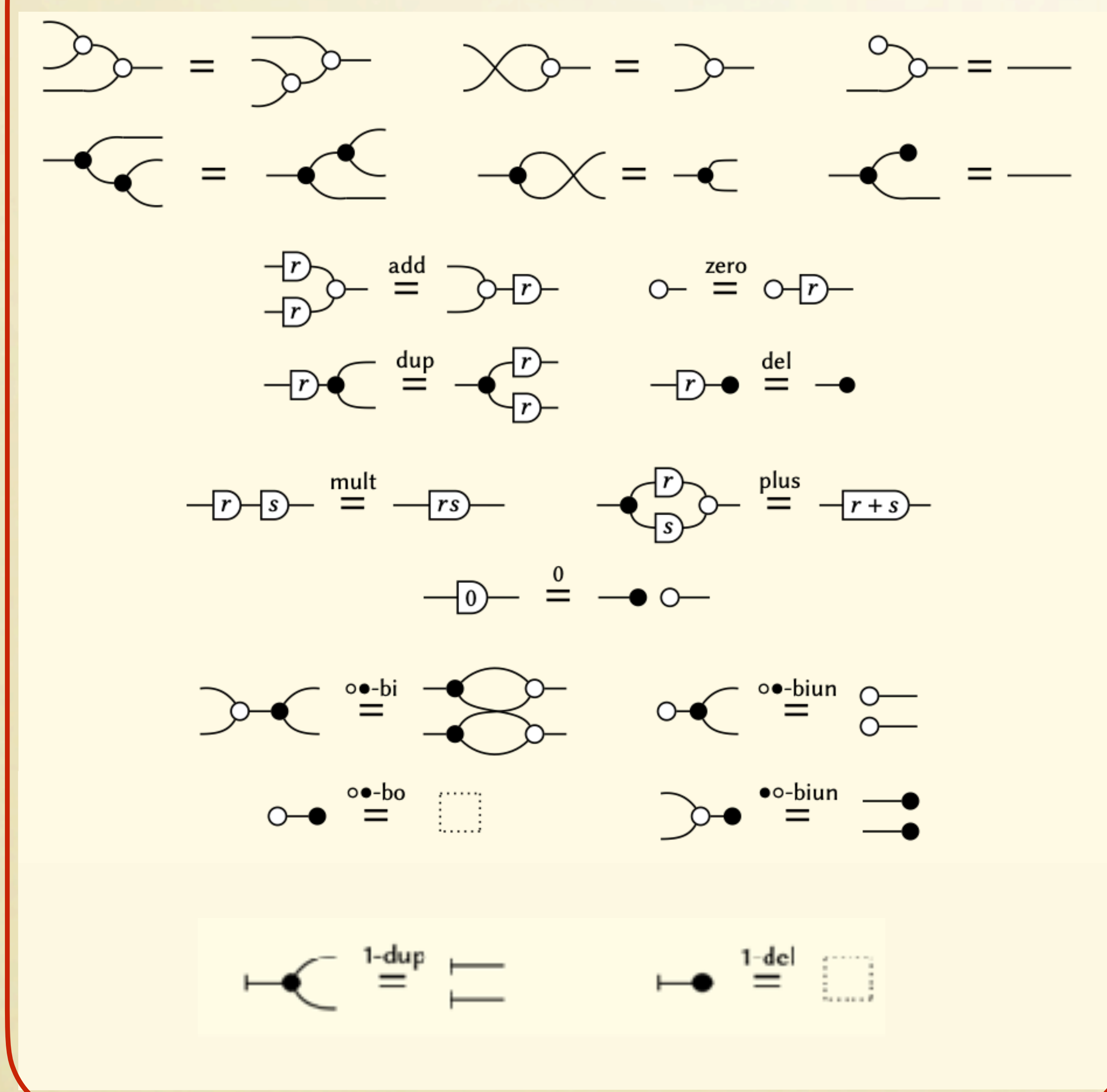
$$f(x) = Ax + \mathcal{N}(\mu, \Sigma)$$

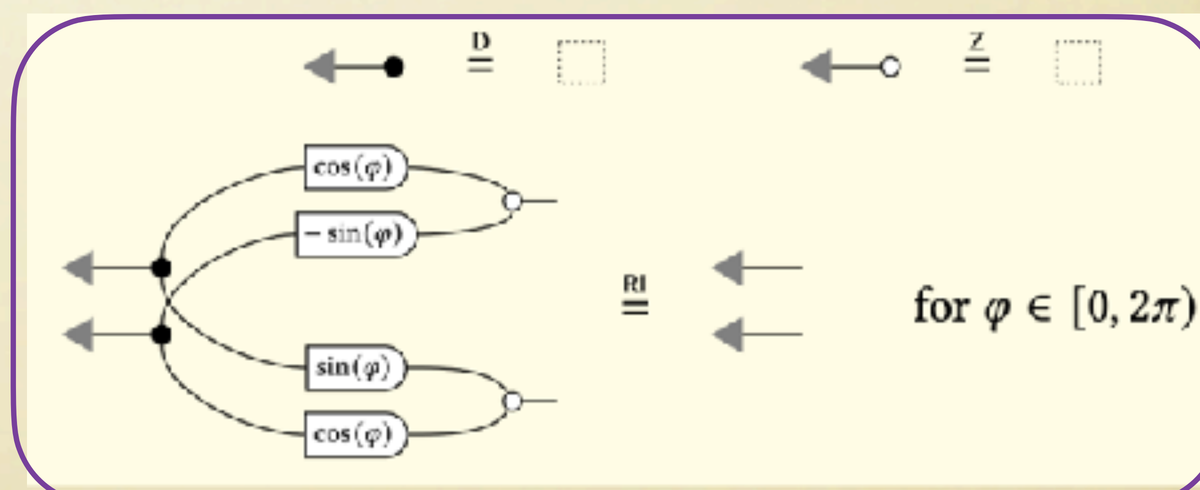
Extended Graphical Quadratic Algebra





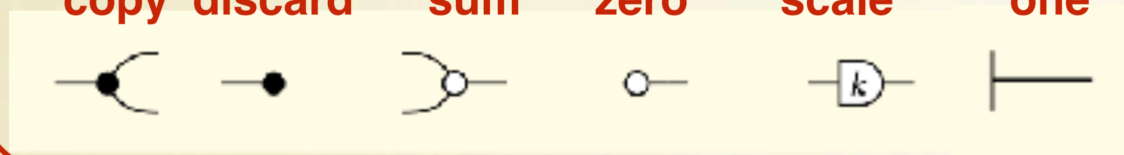
axiomatisation complete for affine maps

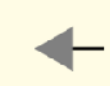


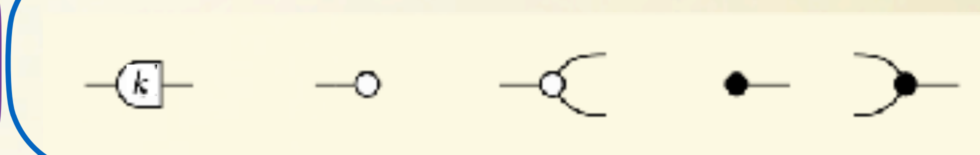


Extended Graphical Quadratic Algebra

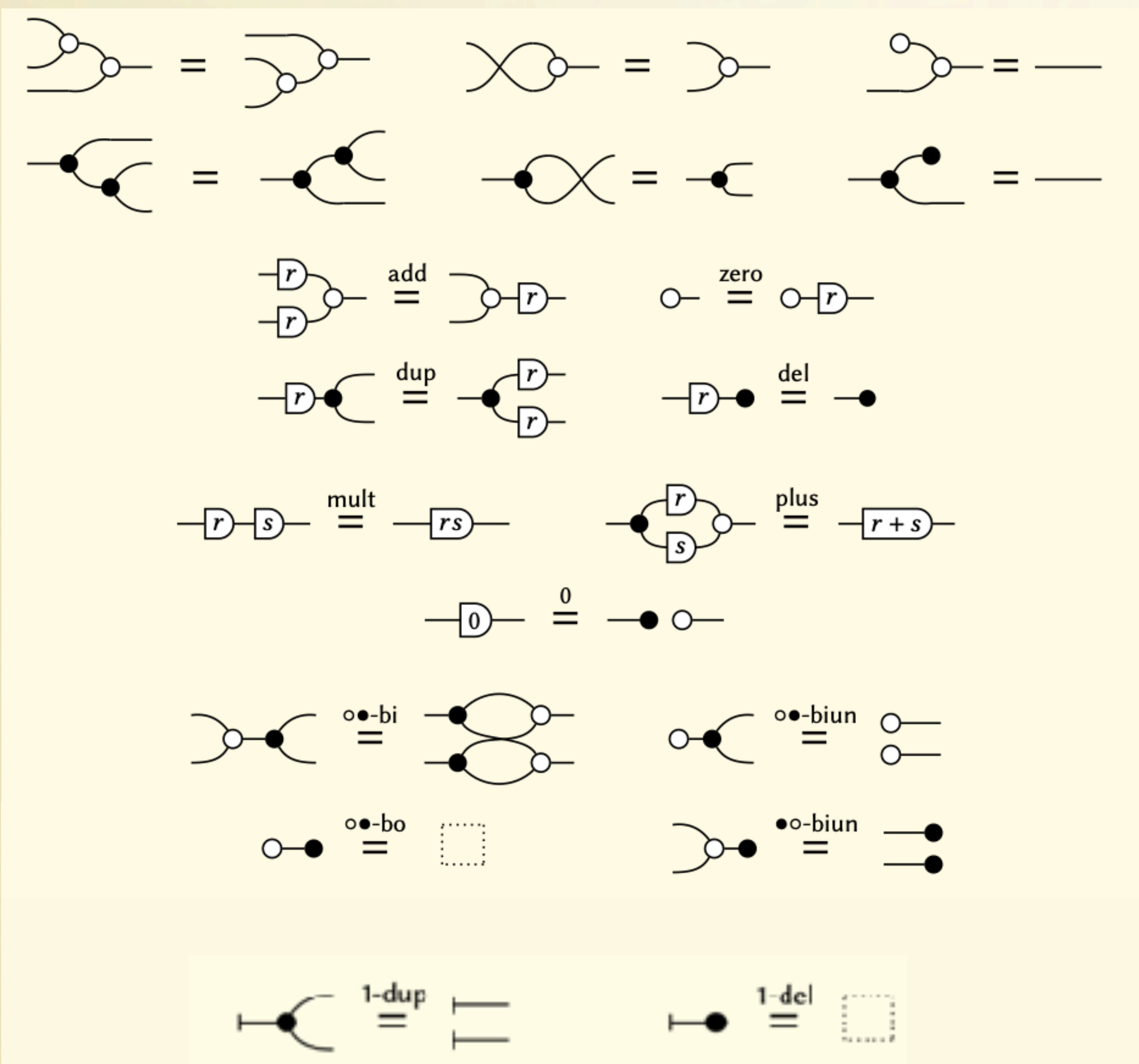
copy discard sum zero scale one



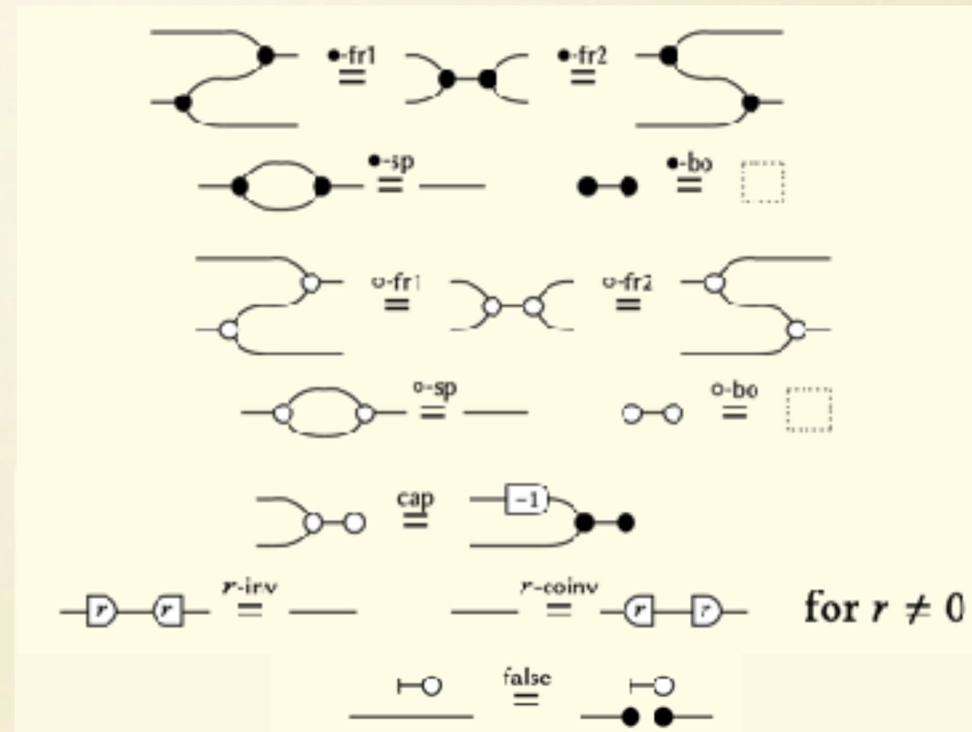


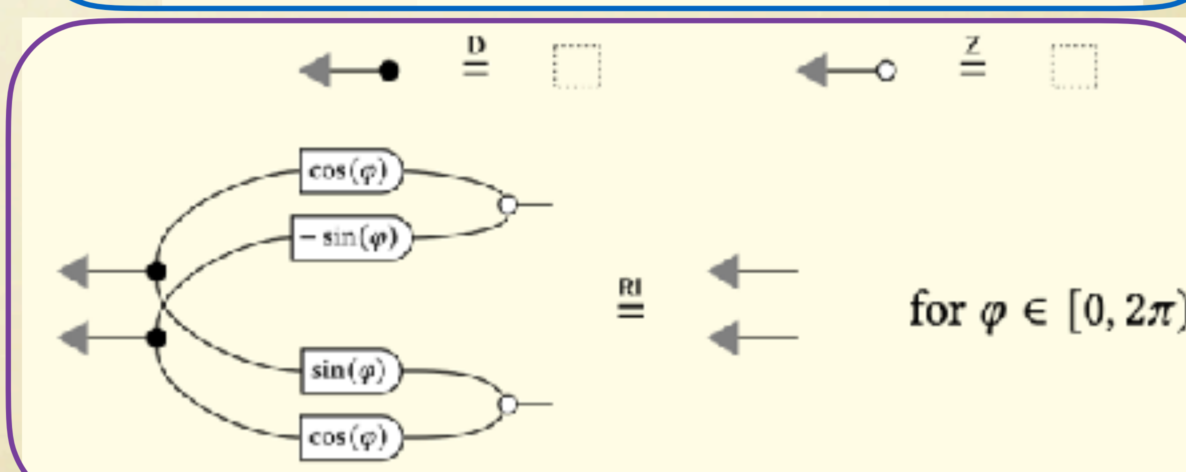


axiomatisation complete for affine maps



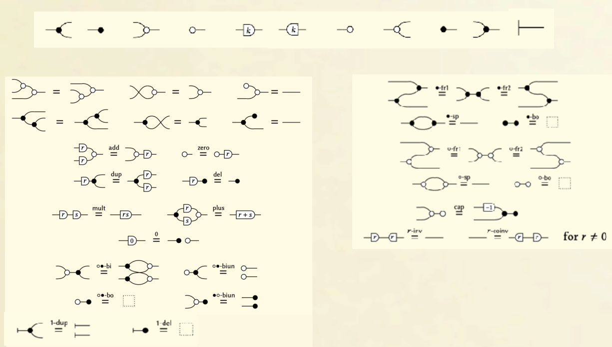
axiomatisation complete for affine subspaces





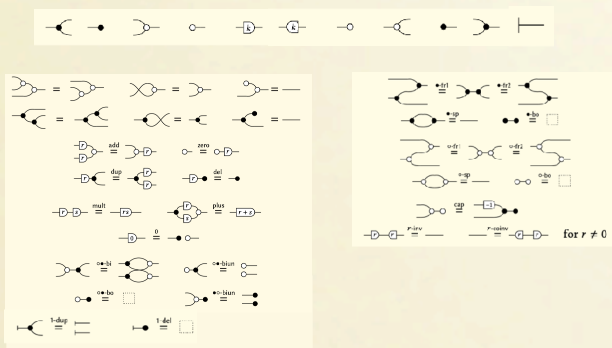
Semantics

Graphical Affine Algebra

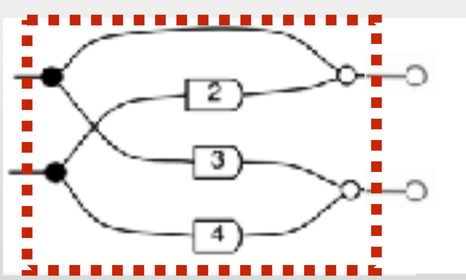


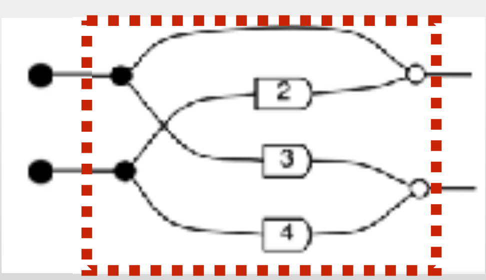
Semantics

Graphical Affine Algebra



Picturing linear subspaces



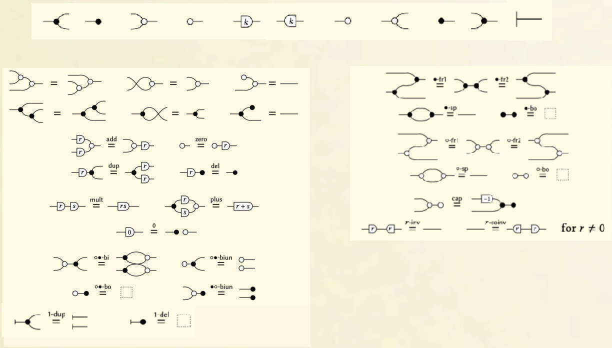


Subspaces as kernels of matrices.

Subspaces as images of matrices.

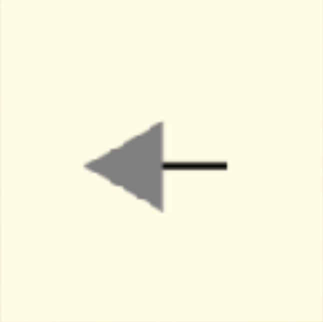
Semantics

Graphical Affine Algebra

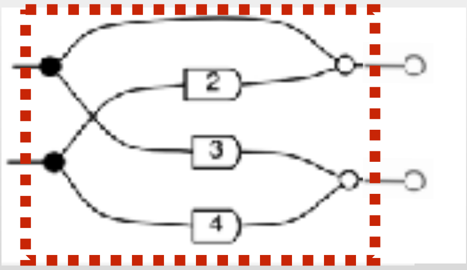


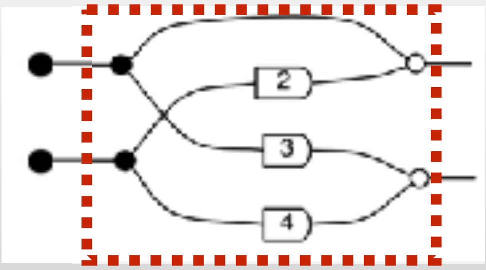
+

Extra generator



Picturing linear subspaces



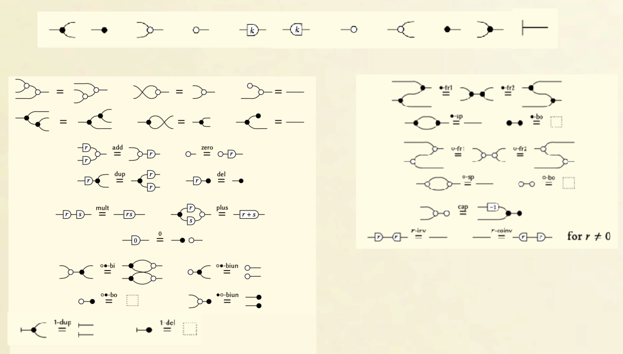


Subspaces as kernels of matrices.

Subspaces as images of matrices.

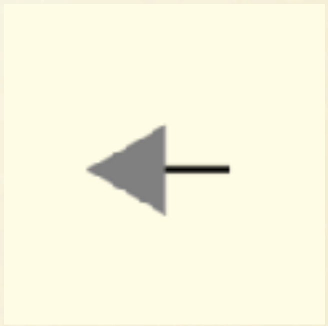
Semantics

Graphical Affine Algebra



+

Extra generator

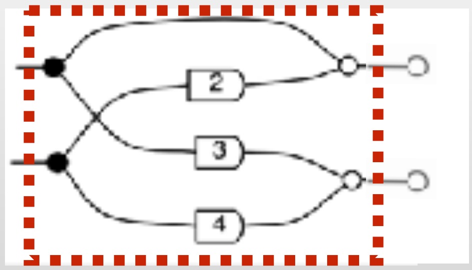


In **QuadRel**

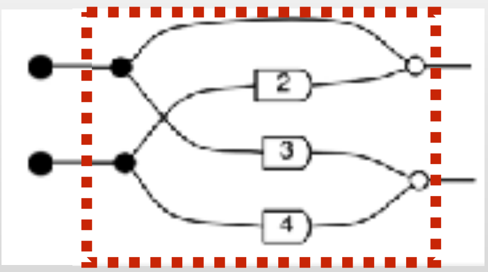
The quadratic function

$$x \mapsto \frac{1}{2}x^2$$

Picturing linear subspaces

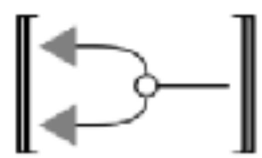


Subspaces as kernels of matrices.

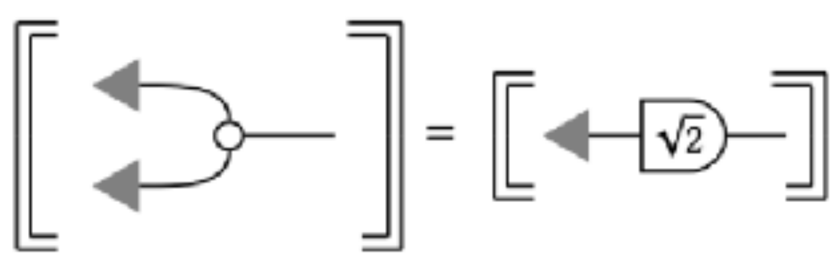


Subspaces as images of matrices.

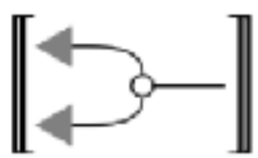
Returning to our example

 $= \; f : y \mapsto \inf\left\{\frac{1}{2}x_1^2 + \frac{1}{2}x_2^2 : x_1 + x_2 = y\right\}$ In **QuadRel**

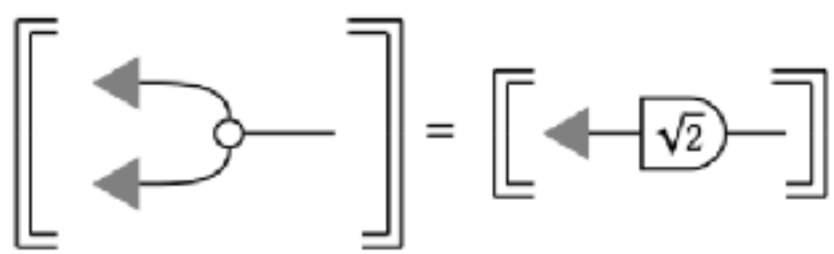
The minimum is attained for $x_1 = x_2 = \dfrac{y}{2}$ so

 $= \; f : y \mapsto \frac{1}{4}y^2$

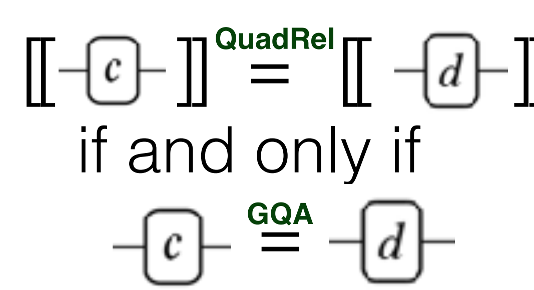
Returning to our example

 $= \quad f: y \mapsto \inf\left\{\frac{1}{2}x_1^2 + \frac{1}{2}x_2^2 : x_1 + x_2 = y\right\}$ In **QuadRel**

The minimum is attained for $x_1 = x_2 = \dfrac{y}{2}$ so

 $= \quad f\colon y \mapsto \dfrac{1}{4}y^2$

Theorem
Graphical Quadratic Algebra is complete for Quadratic Relations.



GQA in Action

Case study: Log-Density

Gauss

$$f(x) = Ax + \mathcal{N}(\mu, \Sigma)$$

Functorial?

QuadRel

Negative Conditional
Log-Density of f

Case study: Log-Density

$$
\begin{tikzcd}
\overrightarrow{\mathrm{GQA}} \arrow[rr, "\textbf{Inductively Defined Functor}"] \arrow[d, leftrightarrow] & & \mathrm{GQA} \arrow[d, leftrightarrow] \\
\textbf{Gauss} \arrow[rr, "\textbf{Functorial?}"] & & \textbf{QuadRel}
\end{tikzcd}
$$

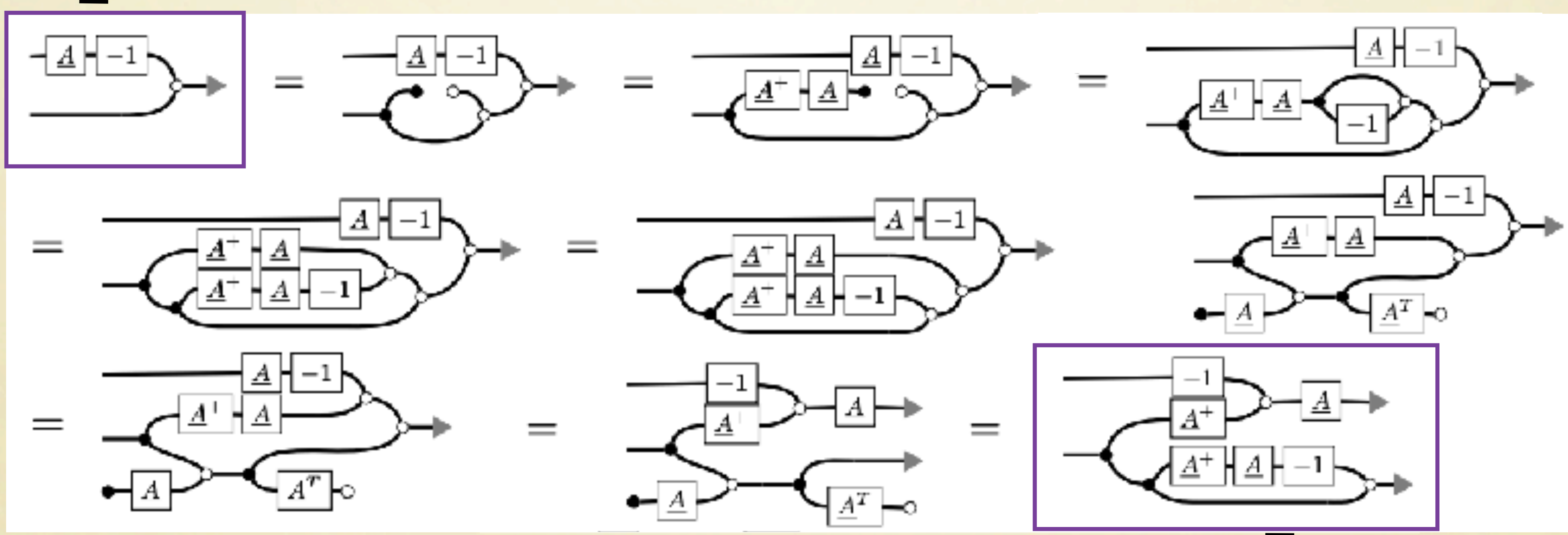
$f(x) = Ax + \mathcal{N}(\mu, \Sigma)$ → Negative Conditional Log-Density of f

Case study: Linear Regression

Ordinary least-squares method: compute x in $Ax = y$ minimising the errors $\frac{1}{2}\|Ax - y\|^2$. Solution is A^+y.

QuadRel

$$\begin{pmatrix} x \\ y \end{pmatrix} \mapsto \frac{1}{2}\|Ax - y\|^2$$



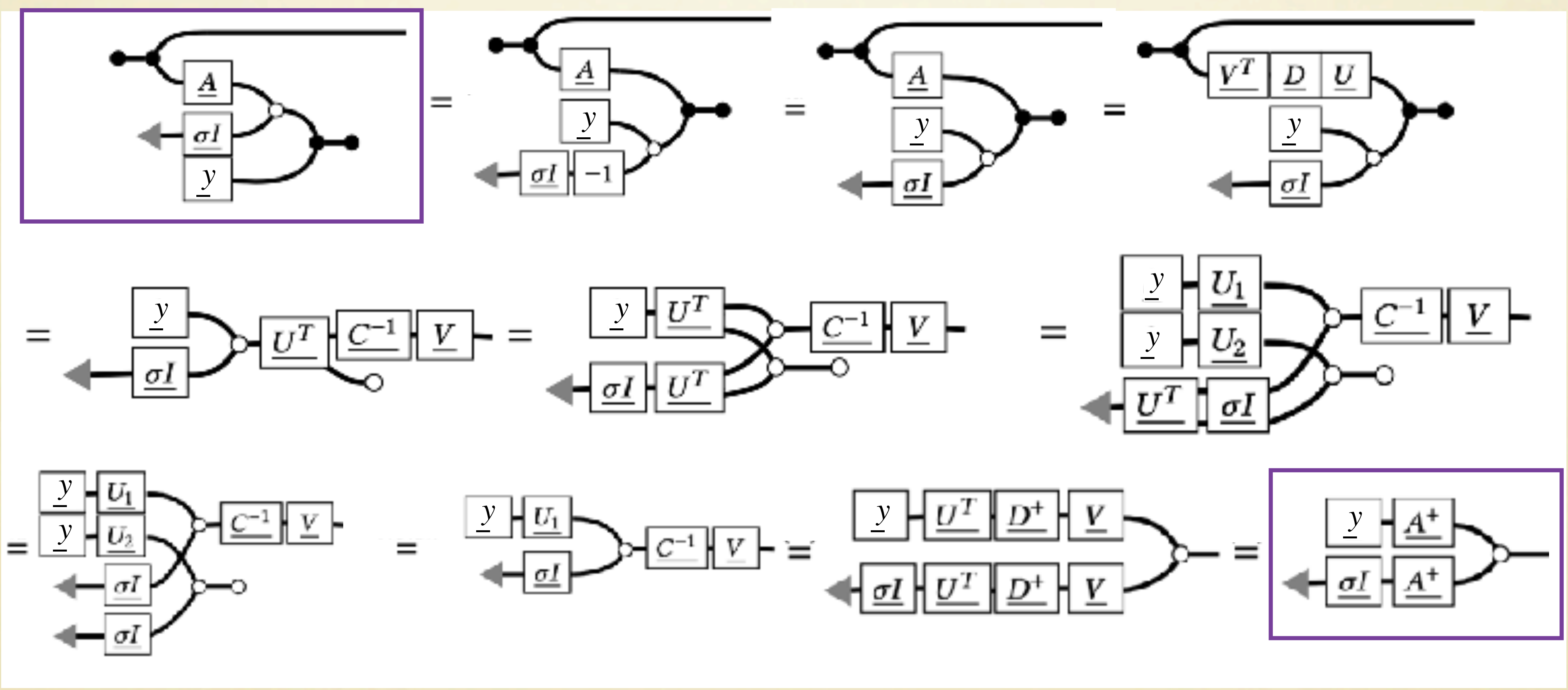
infimum is reached at $\hat{x} = A^+y$

QuadRel

$$\begin{pmatrix} x \\ y \end{pmatrix} \mapsto \frac{1}{2}\left(\|AA^+y - Ax\|^2 + \|y - AA^+y\|^2\right)$$

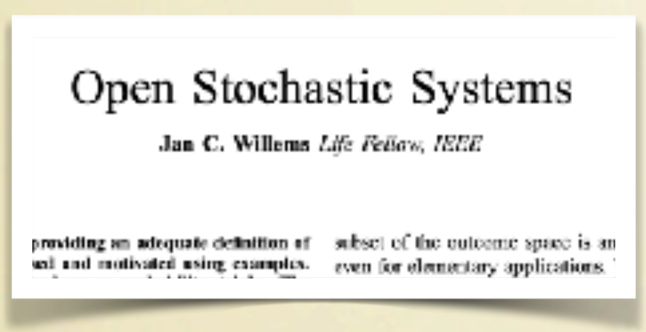
Case study: Linear Regression

Stochastic method: compute random variable x given $Ax + \epsilon = y$, where ϵ is the distribution of errors.
If $\epsilon \sim \mathcal{N}(0\ ,\ \sigma^2 \cdot I)$, x has mean A^+y.

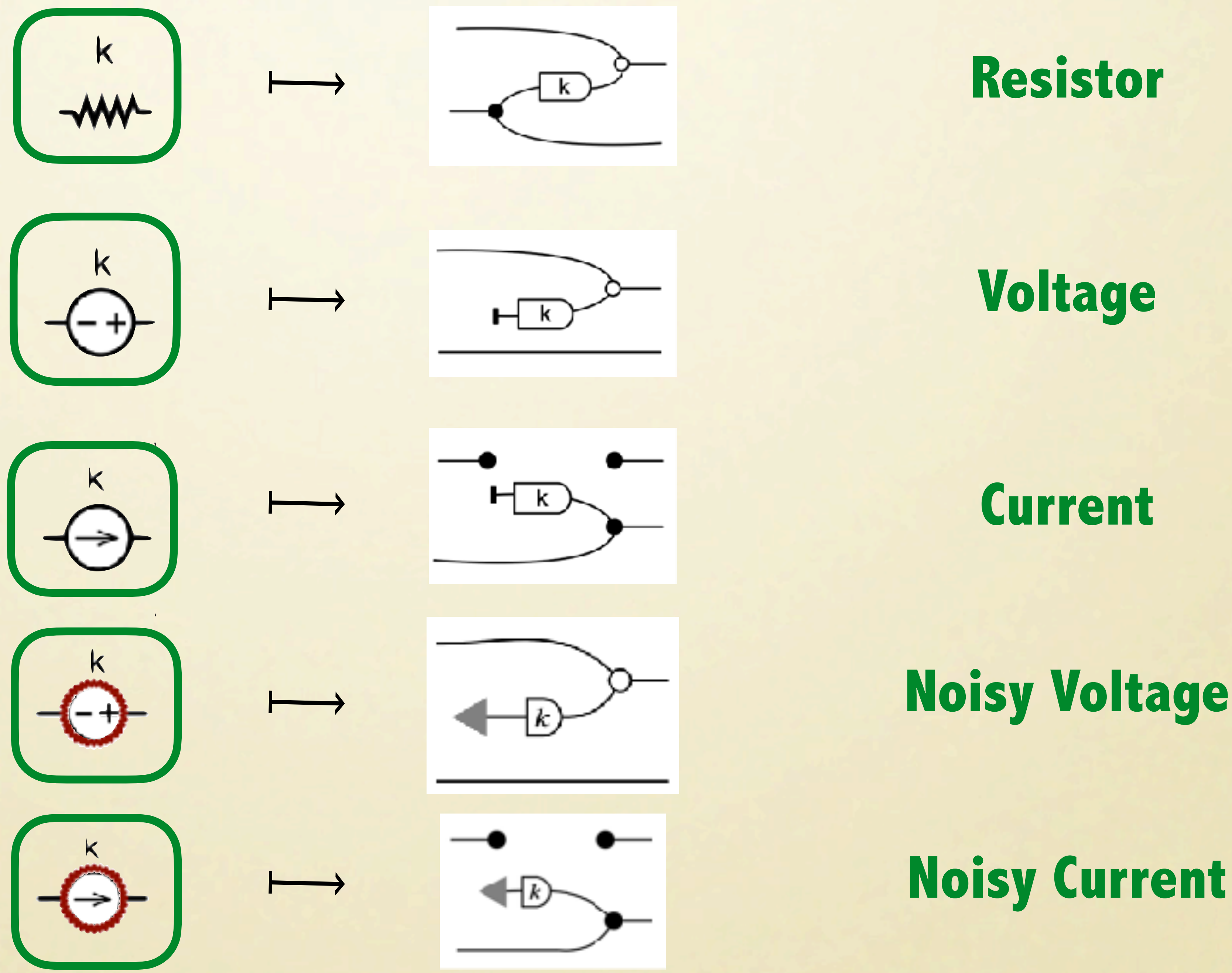


Case study: Noisy Electrical Circuits

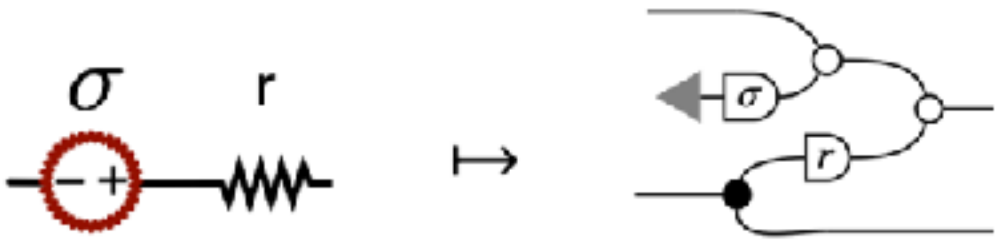
Encode electrical components as string diagrams of GQA

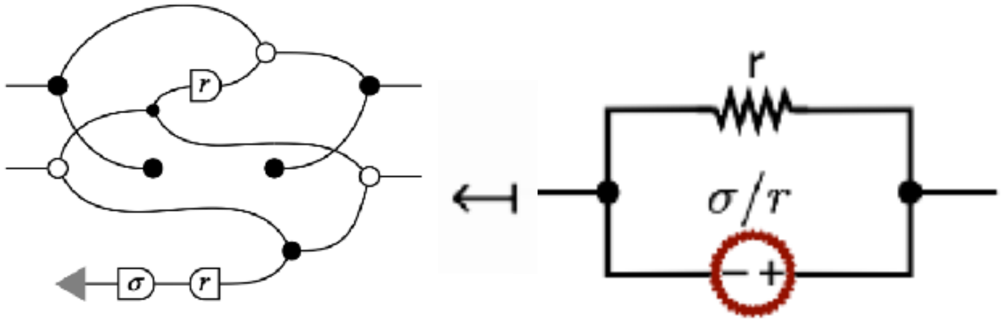

Open Stochastic Systems

Jan C. Willems *Life Fellow, IEEE*

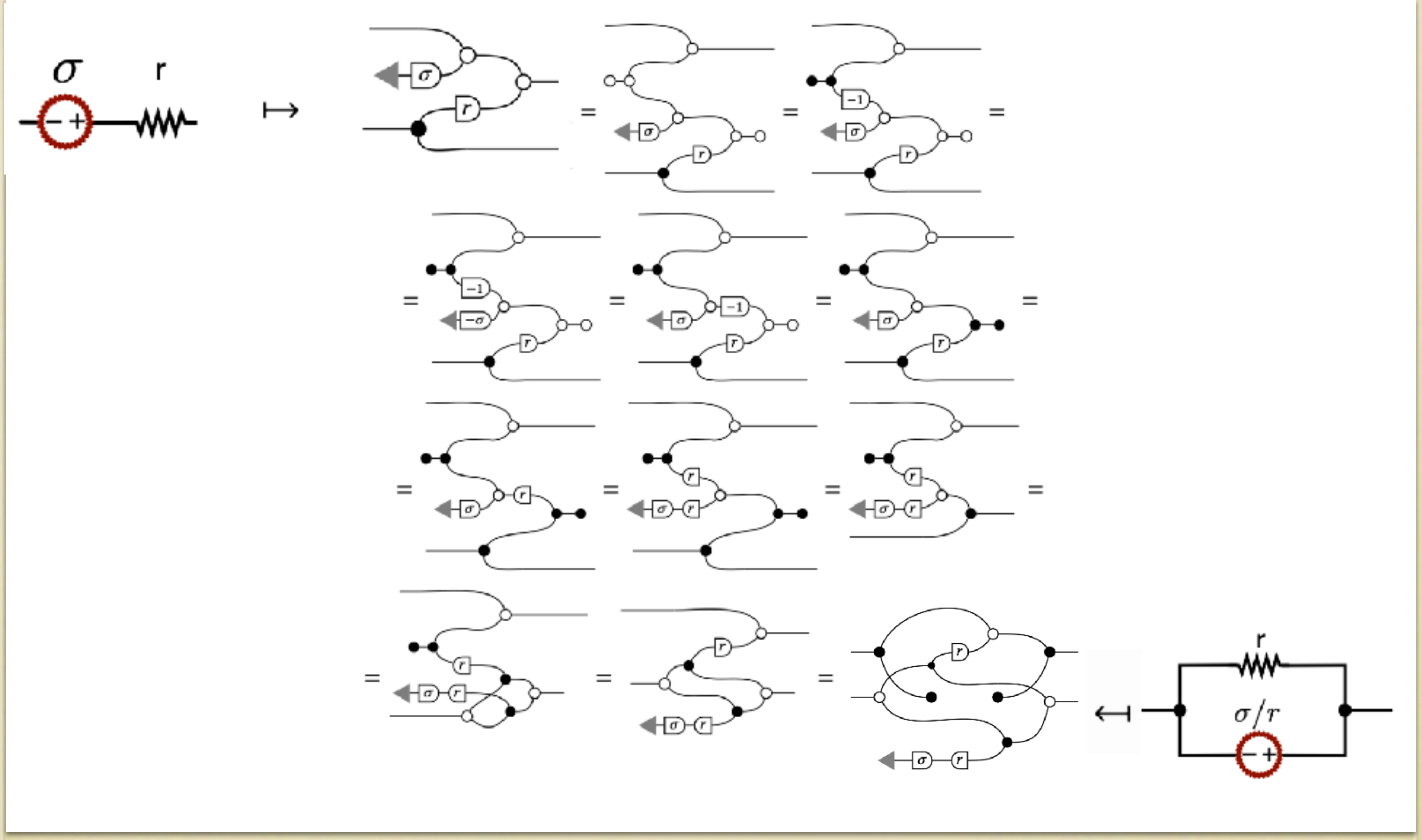


Case study: Noisy Electrical Circuits





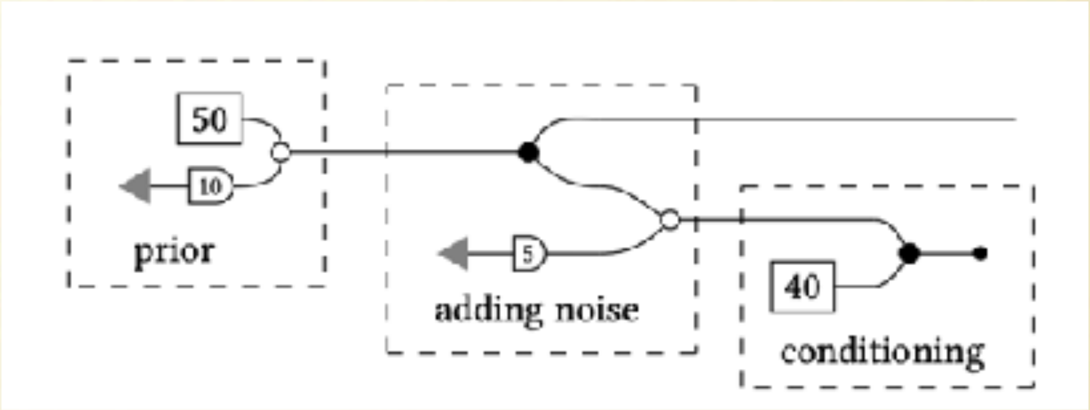
Case study: Noisy Electrical Circuits



Case study: Probabilistic Programming

```
let value = 50 + 10 * normal() in
let measurement = value + 5 * normal() in
measurement =:= 40;
return value
```

$\longmapsto$



Work in Progress:

Decidability Procedure for Contextual Equivalence

Fully Abstract Denotational Semantics

Conclusions

- Graphical Quadratic Algebra is a **complete axiomatic calculus** to reason on quadratic problems **compositionally**.
- Also, (a fragment of) GQA completely captures **Gaussian** probability theory.
- Completeness means we can define **purely syntactic translations** between linear and stochastic perspective.
- Our case studies leave plenty of room for future work, e.g. realisability for **open stochastic systems**, contextual equivalence for **probabilistic programming**.